

The metabolic cost of the febrile host in *Candida auris* and its relatives

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Abstract

Fever exposes pathogenic fungi to temperatures at which growth and respiration need not respond alike, so a pathogen that keeps growing at fever may pay for it in carbon. No fungal pathogen has, to our knowledge, had growth and respiration measured together across the febrile range. Here we recover both rates from 1,200 dissolved-oxygen time series for 20 isolates of *Candida auris* and its relatives *C. haemulonii*, *C. duobushaemulonii* and *C. parapsilosis* at 22 to 44 °C. While cultures grow, respiration keeps rising with temperature as growth turns over, so the apparent carbon-use efficiency peaks at 27 to 34 °C, below body temperature in every group (posterior probability 0.80 to 1.00), and warming from 37 to 40 °C raises respiration per unit growth 1.2- to 1.9-fold in the groups that still grow at 40 °C. *C. auris* retains growth at 40 °C in 58 of 60 wells against 13 of 40 in its relatives, with a growth limit 4.4 °C higher, and pays the smallest fever cost. Once growth is lost, oxygen consumption falls to about a quarter within two degrees but persists. The thermal displacement of *C. auris* is carried by its growth curve, not its respiration.

Introduction

Most fungi cannot grow at mammalian body temperature, and the thermal restriction this imposes is one of the main reasons fungal disease is rare in endotherms^{1,2}. The pathogens that do infect mammals operate at 37 °C, near the upper limit of their growth range, and fever pushes them further toward it. Fever is usually discussed as a growth constraint: a species either grows at 39 to 41 °C or it does not³. But temperature does not act on a single rate. Metabolic rate rises with temperature with a characteristic activation energy⁴, but whether growth and respiration share that sensitivity is an empirical question: across prokaryotes the average sensitivities of growth and of metabolic flux are statistically indistinguishable⁵, whereas growth, unlike respiration, must turn over as the thermal limit is approached. Where respiration keeps rising after growth has begun to fall, the carbon a population respire per unit of biomass it makes must rise with temperature. Growth and yield have been measured against temperature for individual pathogenic yeasts, and in *C. albicans* the yield on glucose falls linearly to zero between the optimum and the maximum growth temperature⁶, but growth and respiration have not been resolved jointly across a febrile thermal series for a pathogen and its relatives, so whether fever imposes a metabolic cost of this kind, and whether that cost differs between a heat-tolerant pathogen and its relatives, is not known.

Candida auris is the natural test case. It emerged on three continents almost simultaneously⁷, in four genetically distinct clades that share 98.7% genome-wide nucleotide identity⁸, and it grows at 39 to 41 °C, temperatures that halt or severely inhibit its closest relatives *C. haemulonii* and *C. duobushaemulonii*³. Thermal adaptation, possibly to a warming environment, has been proposed as the reason it could cross into the mammalian niche when its relatives did not⁹, and the World Health Organization places it in the critical priority group of fungal pathogens¹⁰. Comparisons of growth at fixed high temperatures have established that *C. auris* is the more thermotolerant³, but no study has resolved growth and respiration together, across a thermal series, for enough isolates to separate species, clade and isolate.

Here we do that with dissolved-oxygen respirometry. The dataset is 1,200 oxygen time series (20 isolates × 12 temperatures × 5 replicate wells) for 20 isolates of four *Candida* species at 22 to 44 °C: *C. auris* (12 isolates, three from each of clades I to IV, of the six so far described), its closest relatives *C. haemulonii*

(three isolates) and *C. duobushaemulonii* (two), and the more distant comparison species *C. parapsilosis* (three). Every trace was inspected by hand and given a fit window wherever one existed, and every trace was then re-scored by two independent automatic methods. Isolates are analysed throughout as seven groups: the four *C. auris* clades and the three other species. From each trace a mechanistic oxygen-dynamics model recovers a specific growth rate and an oxygen consumption rate at once¹¹; whether a well grew at all was decided by a per-well likelihood model that sees only time and oxygen; thermal performance curves were fitted hierarchically, isolates nested within groups, in a Bayesian framework; and per-cell respiration and the apparent respiratory carbon-use efficiency ($CUE = G/(G + R)$, apparent because carbon leaving as products other than CO_2 is not measured; Methods) were derived on a matched per-cell basis. Throughout, “growth” in a well means the accelerating oxygen consumption that an expanding population produces, validated against flow-cytometry counts and optical density for bacteria in the method’s original description¹¹ and, for these yeasts, against optical density in an independent plate-reader test (Supplementary Fig. 13). What the oxygen trace supports directly is the divergence between growth and oxygen use across temperature; the carbon interpretation rests on stated conversion assumptions and is treated as conditional on them. Unless stated, values are posterior medians with 95% credible intervals (CrI).

Results

Related taxa frequently leave the growing state near fever

Whether a well grew was decided for every one of the 1,200 oxygen series by a single per-well likelihood model rather than by the manual fit: each trace was fitted with an exponential-to-plateau and a linear-to-plateau mean, both with a free plateau level and Gaussian AR(1) errors, and growth was scored when the likelihood ratio between them exceeded its parametric-bootstrap null at $p < 0.01$ (Methods). The model sees only time and oxygen. It returns a growth rate for every well and leaves no ambiguous category. A blind, pre-registered three-criterion classifier run earlier as an audit agrees with it on 930 of its 931 confirmed-growth wells and is retained as a sensitivity arm (Methods).

Of the 1,200 series, 1,063 show growth, 136 show respiration without detectable growth and one shows no detectable respiration (Fig. 1a). Below 38 °C growth is universal in every taxon. At 40 °C it is retained in 58 of 60 *C. auris* wells (96.7%, isolate-clustered 95% CI 90 to 100) against 13 of 40 in the relatives (32.5%, CI 8 to 60); *C. duobushaemulonii* and two of the three *C. haemulonii* isolates retain no growing well while continuing to consume oxygen (Fig. 1b, c). At the isolate level all 12 *C. auris* isolates and 3 of 8 relatives keep growth in at least half of their wells at 40 °C (risk difference +0.63, 95% CI +0.22 to +0.86, Newcombe hybrid-score method; Fisher’s exact $p = 0.0036$). The fitted growth rate itself falls toward the limit before growth is lost: in the relatives the rate at 38 °C is 13 to 81% of the same isolate’s rate at 34 °C (median 45%), so the transition is a decline that ends in loss rather than a switch (Fig. 1b).

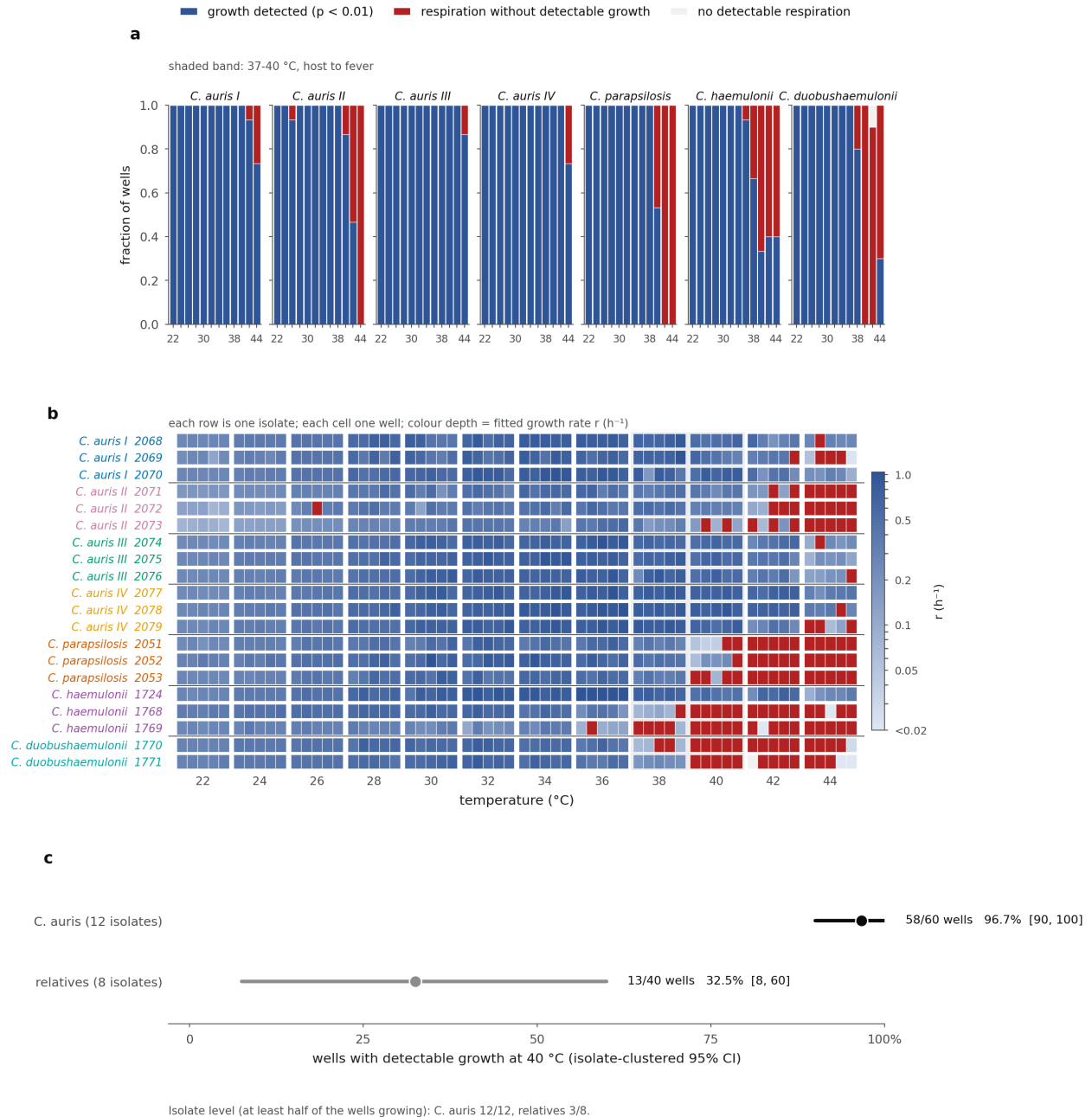


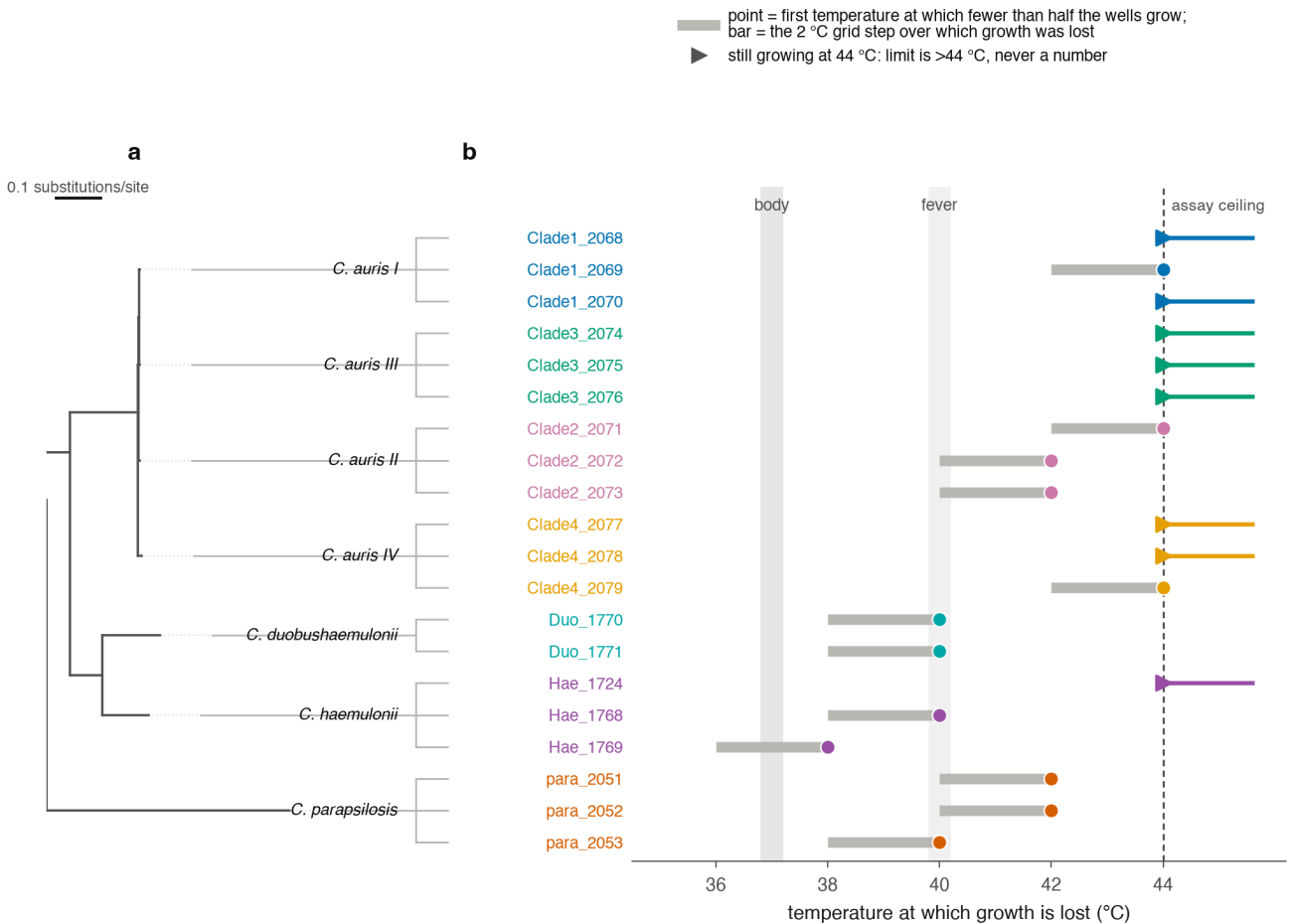
Figure 1 | Related taxa frequently leave the growing state near fever. All 1,200 series scored by the per-well likelihood model. (a) Proportion of wells in each state against temperature, per taxon; shaded band, 37 to 40 °C. (b) Every well: each row is one isolate, each cell one well at one temperature, coloured by the fitted growth rate r where growth was detected and orange where oxygen was consumed without detectable growth. (c) Wells with detectable growth at 40 °C, with isolate-clustered 95% intervals. Orange and the terminal category, *no detectable respiration*, are statements about the oxygen trace under this sealed assay, not about arrest or viability.

High-temperature growth varies sharply among close relatives

High-temperature growth varied sharply among close relatives and was retained far longer across the sampled *C. auris* isolates (Fig. 2). Growth was lost at a median of 40 °C in the relatives, against seven of twelve

88 *C. auris* isolates still growing at the 44 °C assay ceiling (log-rank statistic with exact permutation null,
 89 $p = 0.002$; $p = 0.0004$ excluding Hae_1724). Treating each isolate's limit as interval-censored on the 2
 90 °C grid and right-censored at 44 °C, an interval-censored location model places the *C. auris* limit 4.4 °C
 91 above the relatives' (95% profile CI 2.3 to 7.0 °C; likelihood-ratio $p = 0.0005$; Methods), and the equivalent
 92 proportional-hazards fit gives a hazard of growth loss for *C. auris* 0.15 times that of the relatives (0.03 to
 93 0.43). Within *C. haemulonii* the spread is larger than between many taxa: Hae_1724 retains growth beyond
 94 the assay ceiling while Hae_1769 loses it at 38 °C, and within *C. auris* the three Clade II isolates are the
 95 first to fail, at 42 to 44 °C.

96 As an exploratory estimate, a phylogenetic mixed model on the 20 isolates, with assayed isolates attached
 97 at their group's tip on the reference-genome tree and the eight censored limits multiply imputed (Methods),
 98 attributes about half of the variance in the limit to phylogeny ($h^2 = 0.53$, 0.35 to 0.68 across imputations;
 99 permutation $p = 0.007$). Nearly all of that half is the separation of *C. auris* from the two *haemulonii*-complex
 100 species, since the four *C. auris* clades are almost equidistant on the tree; the other half is variation among
 101 isolates within a clade or species. The estimate rests on seven tips and on where the censored limits are
 102 placed (0.68 or 0.58 if they are set to 44 or 46 °C), so it is reported as a partition of this dataset and not as
 103 a general heritability.



104

105 **Figure 2 | High-temperature growth varied sharply among close relatives and was retained**
 106 **far longer across the sampled *C. auris* isolates.** (a) Maximum-likelihood phylogeny drawn to scale
 107 from 600 concatenated single-copy orthologs of the public reference genomes (one per clade or species; the
 108 assayed isolates were not sequenced), with the branch-length scale shown; tips are connected to the isolate
 109 rows of (b), whose limits come from the assayed isolates. At this scale the four *C. auris* clades are barely
 110 separable: the widest patristic distance inside *C. auris* is 0.012 substitutions per site, against 0.330 to *C.*

haemulonii and 1.146 to *C. parapsilosis*. (b) Temperature at which growth is lost, per isolate: the first measured temperature at which fewer than half of the five wells show growth under the per-well likelihood model, with the loss persisting at every higher temperature. The grey bar is the 2 °C grid step over which growth was lost. Arrowheads mark isolates still growing at 44 °C, whose limit is reported as >44 °C and never assigned a value.

Oxygen consumption persists after growth loss, at reduced and taxon-dependent rates

Whether the respiration-without-growth state of Fig. 1 is metabolically substantial was tested on a fixed, prespecified window (90 to 390 minutes: the 1.5 h thermal-equilibration exclusion plus the 5 h early window already used in this study), applied identically to every well, with traces reaching the oxygen floor treated as right-censored rather than as low metabolism.

C. parapsilosis retained substantial oxygen consumption after growth was lost (2.95 mg L⁻¹ median at 40 °C). *C. haemulonii* (1.05) and *C. duobushaemulonii* (1.22) wells without growth sat only slightly above the 1.0 mg L⁻¹ detection gate. Across the relatives, wells respiring without growth at 40 °C consumed 35 to 45% of what the same isolate’s growing wells consumed at 38 °C (one *C. parapsilosis* isolate, 81%). Because growing wells reach the oxygen floor inside the window far more often than non-growing wells (70% against 0%), their consumption is understated here and the true ratio is lower rather than higher.

The pooled oxygen-weighted share of consumption occurring in growing wells at 40 °C was 0.97 (isolate-level bootstrap 95% CI 0.90 to 1.00) in *C. auris* against 0.63 (0.18 to 0.83) in the relatives, and 0.45 excluding Hae_1724 (Fig. 3). Two isolates cross the species lines: Hae_1724 behaves like a *C. auris* isolate, and Clade2_2073 is the one *C. auris* isolate with wells that consume oxygen at the full rate of a growing well, 7.8 mg L⁻¹ on the window, with no detectable curvature. That isolate is the clearest instance in the dataset of respiration continuing at growth-rate intensity after growth has stopped, and it shows that the contrast is a tendency across sampled isolates rather than a species-level rule.

The rise of respiration with temperature reported below for growing cultures (Fig. 4b) is a property of growing cultures, and it does not continue through the limit. Placing the initial oxygen consumption rate K of every respiring well on an Arrhenius axis within its own isolate, wells at or above that isolate’s growth limit fall to 0.32× the isolate’s sub-limit line (95% CI 0.24 to 0.42; relatives 0.26×, *C. auris* 0.68×), with a steeply negative change of slope (Methods). Two degrees above the limit the median well consumes oxygen at 0.26 times the rate it did two degrees below it. K is a volumetric rate, so part of this fall could be fewer cells rather than less respiration per cell: the growth accrued before the fit window multiplies cell number by e^{rt_s} , which for the wells two degrees below each isolate’s limit is 1.06 to 1.75 (median 1.16; `results/tables/prewindow_growth_factor.csv`), so dividing it out moves the median across-limit ratio only from 0.26 to 0.28. Most of the fall is therefore a fall in respiration itself: growth and respiration decouple across the growing range, and respiration collapses within a grid step once growth is lost. Within a non-growing well that collapse is not a rapid death: over the 15 h window the consumption rate was constant in 72 of the 136 such wells and declined in the other 64, with a median half-life of 7 h, and the rate of decline did not depend on temperature (Methods). Cultures that have stopped growing at fever temperature keep respiring, at a reduced rate, for the duration of the assay.

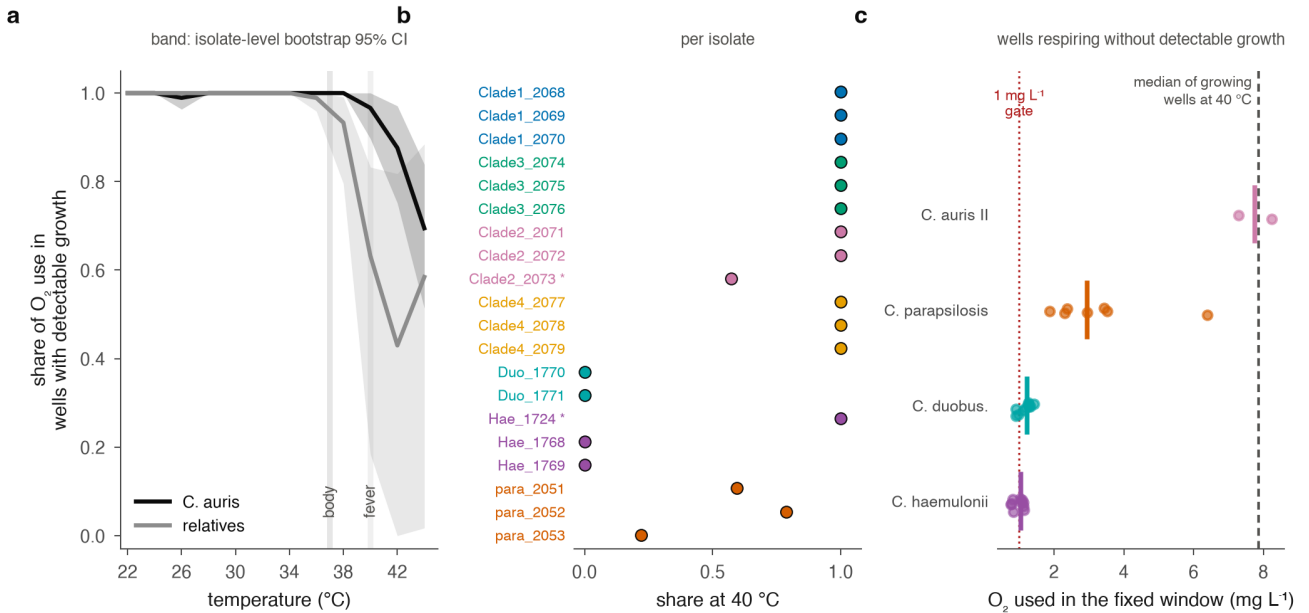


Figure 3 | Oxygen consumption persists after growth is lost, at reduced and taxon-dependent rates. (a) Oxygen-weighted share of the consumption observed on the fixed 90 to 390 min window that occurs in wells with detectable growth, against temperature, pooled for *C. auris* and for the three relatives; band, isolate-level bootstrap 95% CI. Growing wells reach the oxygen floor inside the window far more often than non-growing wells (70% against 0% at 40 °C), so their consumption is understated and the share is a lower bound. (b) The same quantity at 40 °C, per isolate. Asterisks mark the two isolates that cross the species lines, Hae_1724 and Clade2_2073. (c) Oxygen consumed on the fixed 90 to 390 min window by wells respiring without detectable growth at 40 °C; points are wells, the bar the group median, the dotted line the 1 mg L⁻¹ gate used for the no-respiration call, which applies to the total drawdown of the whole trace rather than to this window, so fixed-window values below it are possible; the dashed line is the median of growing wells at the same temperature. *C. auris* clades I, III and IV had no such wells at 40 °C and are absent from (c).

Growth and respiration decouple, placing the apparent CUE optimum below body temperature

The next two sections derive a carbon cost from the same oxygen traces, in a deliberate hierarchy of assumptions. The position of the CUE optimum, the growth-respiration decoupling and the fold-change in their balance across temperature are identifiable from the fitted growth rate r , the volumetric consumption rate K and the time each culture spent growing before its fit window, and need no absolute cell number, carbon quota, oxygen-to-carbon conversion or respiratory quotient. Absolute per-cell magnitudes depend on all of those and carry their uncertainty (Robustness).

Growth rate rose with temperature to a group-level optimum between 31.9 and 36.4 °C and then declined, following the expected unimodal thermal performance curve (Fig. 4a). Per-cell respiration showed no such turnover: it increased monotonically across the range in which each group grew (Fig. 4b; fitted extensions beyond that range are drawn dashed), and leave-one-out cross-validation favoured the monotonic Arrhenius model over a unimodal Sharpe–Schoolfield alternative (Fig. 4b). Growth and respiration are therefore governed by different thermal sensitivities.

This difference is quantified by the activation energies (Fig. 4c). In every group the activation energy of growth exceeded that of respiration: E_G ranged 0.62–1.15 eV and E_R 0.30–0.52 eV. The difference is credibly greater than zero in six of the seven groups (0.36–0.73 eV, 95% CrI excluding zero); in *C. duobushaemulonii*, the least replicated group, the interval includes zero (0.15 eV, −0.06 to 0.36). Growth is thus substantially

more temperature-sensitive than respiration. The decoupling does not depend on the per-cell conversion: comparing the activation energy of the fitted growth rate r directly against that of the volumetric consumption rate K , which is the per-cell respiration under the assumption of no growth before the fit window, gives E_r 0.87–1.23 eV against E_K 0.55–0.62 eV ($P = 1.000$ in each of the four *C. auris* clades and *C. parapsilosis*, and 0.947 for the two remaining relatives pooled), and the same ordering holds under every treatment of the pre-window growth (Robustness). The absolute per-cell respiration activation energies carry a temperature-structured uncertainty from that treatment; the ordering does not.

Because respiration continues to rise while growth saturates and falls, the apparent respiratory carbon-use efficiency ($CUE = G/(G+R)$, with G and R the growth and respiratory carbon fluxes of the same population; Methods) is a unimodal function of temperature that peaks early and then declines (Fig. 5a). The position of that optimum needs fewer assumptions than its level. Writing $G = r \cdot q \cdot N$ and $R = c \cdot K$, where N is the cell density at the start of the fit window, q the carbon per cell and c the oxygen-to-carbon conversion, $CUE = 1/(1 + (c/q) \cdot K/(r \cdot N))$, so the temperature that maximises CUE is the temperature that minimises $K/(r \cdot N)$, and c and q drop out. N is not constant across temperatures: the inoculum grows for the interval t_s between inoculation and the start of the window, so $N = N_{inoc} \cdot e^{rt_s}$ and the quantity to minimise is $K \cdot e^{-rt_s}/r$, in which the common inoculum N_{inoc} cancels but the growth accrued before the window, taken here at the fitted rate r , does not. This model-free estimate, located on the measured temperature grid with isolates resampled before wells (Methods), places the optimum at 27.8, 34.0, 31.0 and 32.0 °C for *C. auris* Clades I to IV and 26.5 °C for *C. parapsilosis*, with $P(T_{opt}(CUE) < 37 \text{ °C}) = 1.00$ for four of the five and 0.94 for Clade IV, whose interval reaches 37.6 °C. For *C. haemulonii* and *C. duobushaemulonii* the grid minimum can be located (31.3 and 28.4 °C) but the intervals are wide (23.9 to 41.6 and 26.6 to 32.3 °C) because growth is lost across much of the upper grid. Omitting the pre-window growth term, as an earlier version of this analysis did, lowers every optimum by 0.4 to 7 °C and is not scale-free.

The parametric (Sharpe–Schoolfield) counterpart on the per-cell scale, which carries the pre-window growth through the inoculum back-projection, gives 29.1–34.3 °C with $P \geq 0.993$ for all seven groups (Fig. 5b), within a degree of the model-free values where both are well constrained. Correcting the back-projection for the vials’ measured equilibration moves the warmest estimate, Clade IV, to 36.6 °C with $P(< 37 \text{ °C}) = 0.80$ (Supplementary Fig. 9). Under every treatment the estimate lies below the growth optimum and below 37 °C in every group; the evidence for the margin is strong in six groups and moderate in *C. auris* Clade IV, the clade that is warmest by every measure in this study.

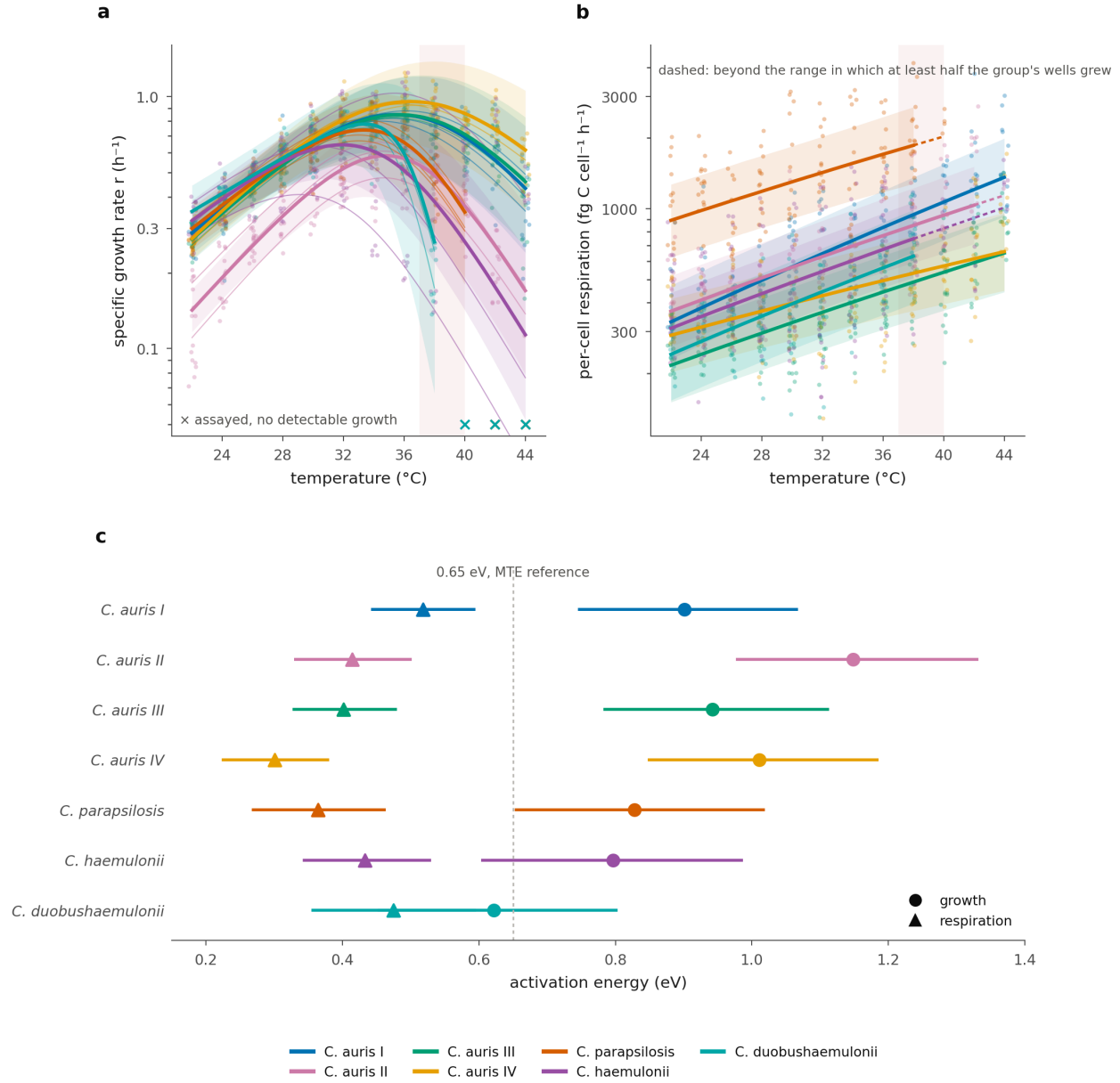


Figure 4 | Growth and respiration decouple with temperature. (a) Specific growth rate r (h^{-1}) and (b) per-cell respiration carbon flux ($\text{fg C cell}^{-1} \text{h}^{-1}$) versus temperature, for all seven groups (points, isolate-level observations; thin lines, per-isolate fits; bold lines, group-level posterior medians; bands, 95% CrI). Crosses in (a) mark temperatures assayed with no detectable growth; in (b) curves are dashed beyond the last temperature at which at least half the group's wells grew. Growth turns over; respiration does not (Arrhenius favoured over Sharpe–Schoolfield by leave-one-out cross-validation). (c) Activation energies of growth (circles) and respiration (triangles) with 95% CrI; dashed line, the 0.65 eV metabolic-theory value. The difference is credible in six of the seven groups; in *C. duobushaemulonii* the interval includes zero.

Fever raises respiration per unit growth, by a taxon-specific amount

This decoupling predicts that warming across the host range should raise respiration per unit growth; we therefore quantified that penalty at body and fever temperatures. Because the economic optimum lies below 37 $^{\circ}\text{C}$, every taxon operates above its cheapest temperature at 37 $^{\circ}\text{C}$ under the assayed conditions. We

express this as the relative respiratory cost, i.e. respiration per unit growth relative to each taxon's own cheapest temperature (Supplementary Fig. 1). At 40 °C (fever), this cost reached 1.32× for *C. auris* Clade IV against 2.95× for *C. parapsilosis*, 3.97× for *C. haemulonii*. For *C. duobushaemulonii*, which retains no detectable growth at 40 °C, the cost is unbounded and no value is reported (Fig. 1).

The three-degree step from body temperature (37 °C) to fever (40 °C) captures the clinical penalty most directly (Fig. 5c). Over this interval *C. parapsilosis* retained 60% of its 37 °C growth rate (0.60, 95% CrI 0.48–0.73) while its respiratory cost per unit growth rose 1.92× (1.56–2.39). The *C. auris* clades were less affected: Clade IV retained 90% of growth (0.90, 0.82–1.00) at a cost increase of 1.24× (1.11–1.37), with Clades I–III intermediate (growth retained 0.69–0.85; cost 1.36×–1.68×). *C. haemulonii* paid 1.95× (1.67–2.34); *C. duobushaemulonii* has no detectable growth at 40 °C, so neither quantity is data-supported for it (Fig. 5c). Fever thus suppresses growth more strongly than respiration; detectable oxygen consumption can persist after growth is lost, and the imbalance is most severe in the relatives. Integrated over a day of constant 40 °C, the fitted growth rate of the median *C. auris* isolate is 83% of its 37 °C value and that of the median relative 56% (a ratio of integrated specific growth rates, not a biomass yield; intermittent-fever profiles in Methods and `results/tables/fever_profiles.csv`).

Differences among the groups. Of the 15 pairwise contrasts in the 40 °C cost among the six groups with measurable growth at 40 °C (*C. duobushaemulonii* excluded from every 40 °C comparison), six resolve (95% CrI excluding a ratio of 1; Supplementary Fig. 4). Five of them place a *C. auris* clade below a relative; the sixth is the one contrast between the two extreme *C. auris* clades, Clade II above Clade IV (1.62, 1.07 to 2.57), the same pair that separates in the displacement analysis (Supplementary Fig. 6). Against *C. parapsilosis* the resolved contrasts are Clade IV (0.45, 0.26 to 0.73) and Clade III (0.54, 0.30 to 0.91); for Clades I and II the interval includes 1 (posterior probability of being cheaper 0.96 and 0.85), so the *C. auris*-versus-*C. parapsilosis* contrast is a species-level tendency rather than a property of all four clades individually. We do not report a finer ranking among the *C. auris* clades, because it depends on the inoculum back-projection (Robustness; Supplementary Fig. 10).

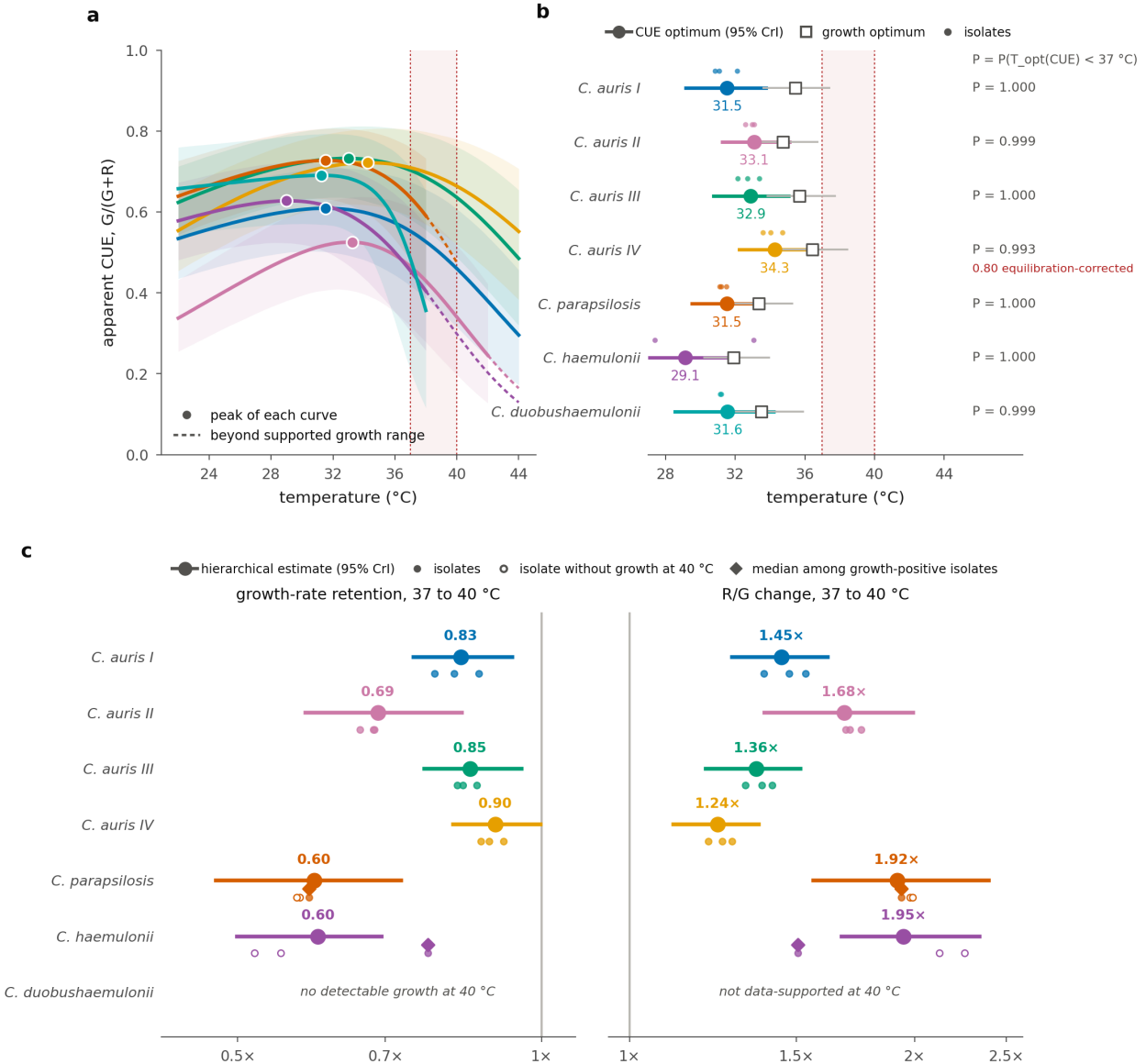


Figure 5 | The apparent carbon-use efficiency peaks below body temperature, and fever raises the respiratory cost of growth by a taxon-specific amount. Panels (a) and (b) are model-derived from the fitted curves; the isolate points in (c) are per-isolate fitted values at the two measured temperatures, and every 40°C quantity requires measurable growth at 40°C . (a) Apparent carbon-use efficiency, $\text{CUE} = G/(G+R)$, against temperature per group; filled points mark each curve's peak; curves are solid over the range in which at least half of the group's wells grew and dashed beyond it, where the fitted curve is an extrapolation; the shaded band spans body to fever temperature. (b) Thermal optimum of CUE (filled points, 95% CrI; open squares, the growth optimum; small points, isolates), with the posterior probability that it lies below 37°C ; for Clade IV the probability after the equilibration correction (Supplementary Fig. 9) is shown in red; shaded band, 37 to 40°C . (c) The 37 to 40°C step: growth retained (left) and the rise in respiratory cost per unit growth (right), as hierarchical estimates with 95% CrI, with isolate values behind them. The *C. haemulonii* estimate at 40°C rests on the one isolate (Hae_1724) that grows there. *C. duobushaemulonii* has no detectable growth at 40°C , so neither quantity is data-supported for it and both are left blank. The relative-cost curves these quantities are read from, the comparative association between growth optimum and fever cost, and the full set of pairwise contrasts are given as Supplementary Figs. 1, 2 and 4.

We next asked whether the variation in fever cost was related to each group’s broader thermal position. Across the six groups with measurable growth at 40 °C, the fitted growth optimum was negatively associated with the cost of a fever (Supplementary Fig. 2): as a posterior over Spearman’s ρ , propagating parameter uncertainty rather than treating the group medians as fixed, $\rho = -0.89$ (95% CrI -1.00 to -0.54 ; $P(\rho < 0) = 1.00$), with *C. parapsilosis* at the cool-optimum, expensive-fever extreme and *C. auris* Clade IV at the warm-optimum, cheap-fever extreme. With six points, four of them clades of one species, and one axis partly geometrically coupled to the growth curve, this is an exploratory association whose sign and not its magnitude is the interpretable quantity; a leave-one-group-out recomputation is given in Supplementary Fig. 3. It is consistent with environmental thermal selection contributing to tolerance of the mammalian host, and it is the pattern the isolate-level analysis below tests directly.

C. auris is displaced along the temperature axis

The fever cost above is a ratio between two chosen temperatures. A more complete question is how far each taxon is displaced along the temperature axis before its carbon economy deteriorates. For every posterior draw we located the temperature minimising respiration per unit growth and solved, on the high-temperature limb, for the crossings at which that ratio reaches 1.5, 2 and 3 times that draw’s own minimum. The ratio is taken from the per-cell curves, so the carbon quota and oxygen-to-carbon conversion cancel in it; the pre-window growth enters through the per-cell respiration fit and is bounded in Robustness.

The ordering of taxa is identical at all three multipliers, Clade IV > Clade III > Clade I > Clade II > *C. parapsilosis* > *C. haemulonii* $\approx$ *C. duobushaemulonii*, and *C. auris* Clade IV reaches any given multiple of its own minimum 6.0 to 7.6 °C warmer than *C. haemulonii* ($P(> 0) = 1.000$ at every multiplier; Supplementary Figs 5 and 6). No multiplier is a biological failure threshold. The 3 $\times$ crossing lies closest to the temperature at which growth is actually lost, at group level and isolate by isolate (Supplementary Fig. 12 and Supplementary Note 2), but because the same fitted growth rate enters both quantities this is a concordance, not an independent prediction: the temperature at which each isolate’s fitted growth rate falls to half its peak predicts the limit equally well (Methods). The threefold rise in respiration per unit growth at the limit therefore describes where growth ends; it is not an energetic threshold that causes it.

Where the displacement comes from can be read off the fitted curves. Swapping one parameter at a time between each *C. auris* clade and each relative, and attributing the change in the 3 $\times$ crossing to that parameter (Shapley decomposition over posterior draws; Methods), the displacement is carried almost entirely by the growth curve’s warm limb: the deactivation energy E_h accounts for 43% of it on average, the deactivation midpoint T_h for 32% and the growth activation energy for 27%, while the respiration activation energy contributes nothing (-1% , and within ± 1 °C in every pair). Against *C. haemulonii*, the closest relative, the midpoint T_h is the largest single term for Clades I, III and IV (2.6 to 2.8 °C of a 4.1 to 7.7 °C displacement). *C. auris* is displaced because its growth declines later and more gradually above its optimum, not because its respiration responds differently to temperature. Above 38 °C the *C. haemulonii* estimate is strongly informed by a single isolate, Hae_1724, and is not a species-level febrile response.

That isolate-level variation is not a trade-off. Across the 20 isolates, a higher growth limit goes with a higher peak growth rate (draw-wise Spearman $\rho = 0.74$, 95% CrI 0.63 to 0.81), a warmer growth optimum (0.85, 0.76 to 0.92) and a smaller 37 to 40 °C cost (-0.85 , -0.90 to -0.78), and the same signs hold within *C. auris* alone (0.57, 0.59 and -0.58) and within the relatives (Methods). The isolates that tolerate the most heat are also the fastest at their optimum and the least penalised by fever: a “hotter is better” pattern rather than a specialist’s price. The peak-rate comparison across species depends on the per-group carbon quota, so the within-*C. auris* value, where the quota is shared, is the cleaner of the two.

Sub-febrile metabolic metrics do not predict high-temperature failure

We asked whether metabolism measured below fever forecasts failure above it. Using only the manual fits restricted to 22 to 38 °C, and validating by leave-one-isolate-out with the complete isolate withheld, growth rate at 38 °C together with the growth optimum predicted the temperature of growth loss with a mean absolute error of 0.93 °C over the twelve isolates whose loss lies inside the assay range; $\log(K/r)$ at 38 °C with its slope gave 1.42 °C, and the two combined 1.10 °C. For the binary outcomes (growth at 40, 42 and 44

°C, and respiration without growth at 40 °C) the held-out log predictive score was highest for the growth-only predictors in every case. Sub-febrile metabolic predictors therefore did not improve out-of-sample prediction beyond growth measurements, and we make no early-warning claim (Supplementary Note 1).

Robustness of the derived quantities

The specific growth rate r and its activation energy are recovered directly from the oxygen dynamics and do not depend on the starting cell number; per-cell respiration, the level of CUE and quantities derived from them inherit the uncertainty of the inoculum back-projection $N_0 = N_{\text{inoc}} \cdot \exp(r \cdot \Delta t)$. We bounded that uncertainty four ways (Supplementary Note 3). Correcting the back-projection for the vials' measured equilibration shifts per-cell respiration by at most +14% near 30 °C and -27% at 44 °C and leaves growth per cell unchanged (Supplementary Figs 7 to 9). Removing the back-projection term altogether, the opposite bound, leaves r , its activation energy and the sub-37 °C CUE optimum unchanged or sharper, and exposes the two results that do depend on it: the fine fever-cost ordering among the *C. auris* clades, whose leader moves from first to fourth, and the between-clade spread in respiration activation energy, which collapses from 0.22 to 0.07 eV (Supplementary Fig. 10); neither is reported as a result. Refitting the growth model with fit windows chosen by the pre-registered classifier's rule instead of by hand moves every group's optimum down by 0.45 to 1.26 °C with every interval including zero (Supplementary Fig. 11). Because r is fitted while oxygen falls, every well entering the primary fits was refitted on its manual window truncated where measured oxygen first fell below half its value at the window start; the manual windows already end near that level (median 0.51 of the start), so the truncation touched 592 of 1,222 windows by a median 6% and moved the CUE optimum by less than 0.1 °C in every group (Supplementary Note 4).

What no treatment of N_0 can change is the algebra: the position of the CUE optimum, the growth-respiration decoupling (E_r against E_K) and the fold-change in the respiration-to-growth balance across the range ($3.6\times$ to $10.5\times$ with the pre-window growth term, $3.1\times$ to $6.5\times$ without it) are identifiable from the fitted r and K and the pre-window interval alone, verified against the pipeline's own identities across the 1,055 growing series that carry an accepted fit window. The absolute inoculum, carbon quota, oxygen-to-carbon conversion and respiratory quotient enter only the level of CUE and per-cell respiration: because $\text{CUE} = 1/(1 + x)$ with $x = K/(rN)$ and the combined conversion factor, a temperature-independent change in μ moves the level of CUE nonlinearly and changes the amplitude of the curve on the CUE scale, but leaves the temperature of its optimum unchanged. Rankings among groups are preserved by any change in the factors shared across groups (the respiratory quotient and the carbon density), because $\text{CUE}_A > \text{CUE}_B$ if and only if $x_A/q_A < x_B/q_B$ whatever the shared factor; they are not preserved by a change in the per-group carbon quota, which enters only the *C. parapsilosis* comparisons since the other six groups share one quota. The one assumption the primary conclusions do carry is the rate of growth before the fit window, and the three treatments bound its effect: the optimum moves by up to 1.6 °C and, for the warmest clade, its posterior probability of lying below 37 °C by 0.80 to 1.00.

Discussion

The central contrast in these data is between retaining growth at fever and paying more oxygen for it. Across the *Candida* taxa tested, respiration rises with temperature through the range in which growth turns over, so every group that grows at body and fever temperature pays more oxygen per unit growth there than at its own optimum. *C. auris* keeps growing across the febrile range and, in seven of twelve isolates, beyond the assay ceiling, and its warmest clades pay the smallest fever cost; in most of the sampled relatives growth is lost near 40 °C while oxygen consumption persists at reduced, taxon-dependent rates. The contrast is an isolate-level risk difference of +0.63 (+0.22 to +0.86) at 40 °C and a limit 4.4 °C (2.3 to 7.0) higher. That *C. auris* tolerates more heat than the *C. haemulonii* complex was known from growth at fixed temperatures³; what the thermal series adds is that the difference is not uniform within a species, that it is a decline ending in loss rather than a switch, and that it is a property of growth and not of respiration. One *C. haemulonii* isolate grows beyond the assay ceiling, the three *C. auris* Clade II isolates are the first of their species to fail, and, in an exploratory partition, about half of the isolate-level variation in the limit is not accounted for by phylogeny.

The measured result on carbon is a divergence: growth turns over below body temperature while respiration keeps rising, so oxygen consumed per unit of growth is lowest at 27 to 34 °C in every group and rises through the host range. Turning that ratio into an apparent carbon-use efficiency requires a carbon quota, an oxygen-to-carbon conversion and a respiratory quotient, all of which are stated in Methods and none of which was measured across temperature here. The position of the optimum survives every treatment of those quantities, because a temperature-independent conversion cannot move it; only its level and the finer ranking among the *C. auris* clades depend on them, and the finer ranking is not reported. The evidence that the optimum lies below 37 °C is strong in six groups ($P \geq 0.993$ under the fitted curves; 1.00 under the model-free estimate for three *C. auris* clades and *C. parapsilosis*) and moderate in *C. auris* Clade IV, the clade closest to body temperature by every measure, where the equilibration-corrected probability is 0.80 and the model-free interval reaches 37.6 °C; for *C. haemulonii* the model-free interval reaches 41.6 °C because growth is lost across much of the upper grid. The claim we make is therefore about the assayed conditions: in this medium, across these temperatures, body temperature lies above the carbon-efficient range of every taxon measured, and fever widens the gap, least in the warmest *C. auris* clades. Whether the same holds in host tissue, where the carbon source, oxygen tension and immune pressure all differ, is the implication to test, not a result of this study.

The displacement of *C. auris* along the temperature axis is a property of growth: decomposed parameter by parameter, it comes from a later and shallower decline of growth above the optimum, with no contribution from the temperature response of respiration, and no fitted quantity predicts an isolate's limit better than its own growth curve. It is not paid for: across isolates, and within *C. auris* alone, a higher limit goes with a faster peak and a smaller fever cost. Respiration sets the cost of the febrile host while growth continues, and collapses within a grid step once it stops.

Three limitations bound these conclusions. First, growth is inferred from the oxygen trace, and the inference was checked rather than assumed. In an independent optical-density test of four isolates at 37, 40 and 42 °C (Supplementary Fig. 13), every isolate-temperature combination the likelihood model scored as growth grew, the one it scored as no growth at 40 and 42 °C (Hae_1768) rose at most 1.2- to 1.8-fold, a single early doubling, beside wells of *C. auris* that rose 20-fold, and the rank order of the four isolates was the same by both methods at every temperature. The two combinations at the edge of the vial limit (*C. parapsilosis* and Clade II at 42 °C) grew slowly in the aerobic plate where the sealed vial registered nothing, so “no detectable growth” in a vial means no growth fast enough to curve the oxygen trace within 15 h under sealed conditions, and slow aerobic growth can persist one grid step beyond that limit. The species contrast and the fever-cost ordering are unaffected; the isolate-level limits are conservative by up to one grid step for the isolates nearest them. Second, the carbon balance is not closed. The species are Crabtree-negative and *C. auris* has been reported not to excrete ethanol, glycerol or acetate under the conditions tested (Methods), but that has not been shown across 22 to 44 °C in this medium, and CO₂, substrate and extracellular products were not measured; the apparent CUE is an upper bound on the true efficiency wherever such products form. Third, all measurements come from one laboratory medium at set temperatures, with an instantaneous thermal response assumed in the fever-profile integrals, so the host and fever numbers are the metabolic penalty of temperature under assayed conditions and not a measurement of infection.

What remains unresolved is the basis of the differences among the *C. auris* clades. The four clades differ up to twofold in peak growth rate, yet their reference proteomes are 99.4 to 100% identical at the amino-acid level (Methods), the assayed isolates were not sequenced, and a public clade-resolved transcriptome at a controlled condition (Clade I against Clade II at 37 °C; GEO GSE165762) shows no coordinated shift in central-metabolic enzyme transcripts (Wilcoxon $P = 0.93$; Methods). Thermal position also varies among isolates within a clade as much as between clades, which is what a trait varying at the isolate level would produce. Resolving it will require absolute, spike-in-calibrated proteomics under these thermal conditions, together with a panel large enough to treat thermal position as a quantitative trait.

What this study measures is a separation between growth and respiration along the temperature axis, and where each taxon sits on it: *C. auris*, which keeps growing furthest into the febrile range, pays the least for it. Whether that oxygen cost is a carbon cost in the host is the conditional interpretation this paper sets out, and the experiment it points to.

Methods

Isolates and culture. Twenty isolates were assayed: twelve *C. auris* (three from each of clades I–IV), three *C. parapsilosis*, three *C. haemulonii* and two *C. duobushaemulonii*, obtained as identified strains from the MRC Centre for Medical Mycology, University of Exeter. The four-digit number in each isolate name is its collection number; species and *C. auris* clade assignments are the Centre’s and, since the assayed isolates were not sequenced here, were not independently verified. Cultures were grown in YMS medium (4 g L⁻¹ yeast extract, 4 g L⁻¹ malt extract, 4 g L⁻¹ sucrose) and inoculated at 1.5×10^7 cells L⁻¹, counted by haemocytometer for every culture. All four species are Crabtree-negative, so under aerobic conditions oxygen consumption tracks their carbon catabolism; a Crabtree-positive species would ferment even with oxygen present.

Experimental design. The dataset is 1,200 oxygen series (20 isolates $\times$ 12 temperatures $\times$ 5 replicate wells); of the 1,063 wells scored as growing by the likelihood model, 1,055 carry an accepted manual fit window and enter the thermal performance fits, and the other eight are counted in Fig. 1 only. The five wells of an isolate at a temperature are five biological replicates, each inoculated from its own overnight culture. Each 24-vial SensorDish held one group at one temperature (three isolates $\times$ five replicates, two isolates for *C. duobushaemulonii*, plus blanks). Six SensorDish readers, each in its own incubator, ran in parallel, and each reader was used at the same pair of temperatures throughout the study (22 and 24, 26 and 28, 30 and 32, 34 and 36, 38 and 40, 42 and 44 °C, one temperature per day), so a group’s twelve temperatures were covered on two consecutive days and every group met the same reader and incubator at a given temperature. Groups were run in succession: the four *C. auris* clades on 18 to 25 June 2026, *C. parapsilosis* on 27 to 28 June, *C. haemulonii* on 14 to 15 August and *C. duobushaemulonii* on 21 to 22 August (results/tables/run_log.csv, from the reader exports). Reader and incubator are therefore confounded with temperature pair, identically for all groups, so that every between-group comparison at a given temperature is made on the same reader; group is confounded with run date, and a run-date effect cannot be excluded by design, although *C. parapsilosis* (June) loses growth with the August relatives, Hae_1724 (August) retains it with *C. auris*, and the September optical-density plates reproduce the ordering of the isolates tested. Isolates are treated as the unit of independent replication throughout (isolate-clustered intervals, isolate-first resampling).

Oxygen over the fit window. Crabtree-negative yeasts can ferment once oxygen becomes limiting, so the rate quantities used for the thermal performance curves and the carbon cost are early-trace rates: K is the consumption rate at the start of the fit window, and r is fitted over the window during which oxygen falls from that level toward the sensor floor. Measured dissolved oxygen at the window start was 6.5 to 12.9 mg L⁻¹ over the 1,055 windows entering the fits (median 9.4; median by setpoint 10.7 mg L⁻¹ at 22 °C falling to 8.6 at 44 °C), and at the window end a median 0.51 of the start value. These are the concentrations reported by the PreSens software, which converts the sensor’s percentage air saturation to mg L⁻¹ with its stored calibration, pressure and salinity settings; they run 20 to 40% above the tabulated air-saturation concentration of fresh water at each setpoint, so the absolute oxygen scale is offset by a factor common to every well. A common factor on oxygen multiplies K, and hence per-cell respiration, by the same factor in every well and cancels from the CUE optimum, the cost ratios and the activation energies; it enters only the absolute level of per-cell respiration, which is already treated as approximate. Oxygen limitation late in a window remains a possible influence on r that the model does not remove (tested under High-oxygen refill below), and the carbon-use efficiency is called apparent for that reason among others. The fixed-window drawdown of Fig. 3, by contrast, is the oxygen consumed over 90 to 390 min and includes wells that reach the floor inside the window (Oxygen consumption after growth loss, below).

Respirometry. Dissolved O₂ concentration was recorded continuously with PreSens SensorDish readers across a 12-point temperature series (22–44 °C in 2 °C steps), with five replicate wells per isolate and temperature. Vials are inoculated near bench temperature and reach setpoint over tens of minutes; the first 1.5 h of every trace is excluded from fitting on that basis, and the internal vial temperature was logged continuously and used in the equilibration sensitivity analysis (Supplementary Fig. 7).

Growth and respiration from the oxygen trace. Specific growth rate and volumetric O₂ consumption rate were recovered simultaneously from each series using the mechanistic oxygen-dynamics model of ref. 11,

which couples exponential population growth with biomass-proportional oxygen consumption: $O_2(t) = O_{2,0} + (K/r)(1 - e^{rt})$, where r is the specific growth rate (h^{-1}) and K the volumetric O_2 consumption rate at the window start. Curves were fitted by Levenberg–Marquardt least squares (`minpack.lm::nlsLM`). The interval of each trace used for fitting was selected manually, by inspection of the trace, because no automatic rule handles equilibration transients, plateaus and occasional sensor drift as reliably. Two roles are kept apart throughout: manual judgement selects the window in which a rate is estimated for the thermal performance curves, and never decides whether a rate exists. Every statement about whether a well grew comes from the per-well likelihood model described below, which sees only time and oxygen and has no access to the manual decisions. The sensitivity of the fitted curves to the window rule is reported under Robustness of the derived quantities (Supplementary Fig. 11). Per-cell respiration, carbon-use efficiency and derived quantities were computed as in that study.

Per-cell scaling. Growth and respiratory carbon flux were expressed on a matched per-cell basis, so that the apparent respiratory carbon-use efficiency could be computed as $CUE = G/(G + R)$. It is apparent in two senses: carbon leaving the cell as products other than CO_2 is not measured, so $G/(G + R)$ is an upper bound on the true efficiency where such products form, and the oxygen-to-carbon conversion is taken as temperature-independent. The taxa reported here are Crabtree-negative and *C. auris* has been reported not to excrete ethanol, glycerol or acetate under aerobic growth, but that has not been shown for every species across 22–44 °C in this medium. The per-cell carbon quota is taken as a single temperature-independent value, and it is that assumption which makes the per-cell activation energy of growth directly comparable with that of respiration. Cell volumes were $8 \mu m^3$ for the *C. auris* clades, *C. haemulonii* and *C. duobushaemulonii* (from cells of 2–3 μm measured here for *C. auris*, within the range given in the species description¹², and applied unchanged to the two *haemulonii*-complex species, whose cells are of the same size class) and $32 \mu m^3$ for *C. parapsilosis* (a prolate ellipsoid of $3.25 \times 5.75 \mu m$, the midpoints of the published range of $2.5\text{--}4 \times 2.5\text{--}9 \mu m^{13}$), converted to carbon at $165 \text{ fg C } \mu m^{-3}$, from a dry-mass density of $0.38 \text{ pg } \mu m^{-3}$ (34% dry weight at 1.11 g cm^{-314}) and a carbon fraction of dry mass of 0.44, from the elemental composition of yeast biomass¹⁵: a carbon quota of 1,320 fg C per cell for six of the seven groups and 5,280 fg C for *C. parapsilosis* (`results/tables/group_quota.csv`). Oxygen was converted to respired carbon at a respiratory quotient of 1 (12/32 g C per g O_2), the same value for every group. The assumption is not innocuous across 22–44 °C: fungal cell volume and wall composition can shift under thermal stress, and any systematic temperature dependence in the carbon quota would propagate into the per-cell growth activation energy rather than cancelling from it. The scale-free comparisons reported in the Results avoid the assumption altogether.

Hierarchical thermal performance models. Growth was modelled with the high-temperature-inactivation form of the Sharpe–Schoolfield function¹⁶ and per-cell respiration with a Boltzmann–Arrhenius function⁴, both fitted hierarchically with isolates nested within groups in a Bayesian framework using `brms`¹⁷ on Stan¹⁸, with the per-series standard error carried as a measurement-error term. Sampling used four chains of 4,000 iterations (1,000 warmup), `adapt_delta` = 0.99 and a fixed seed. The thermal optimum of a fitted curve follows from its parameters as $T_{opt} = 1/(1/T_h - \text{logit}(E/E_h)/(E_h \cdot k^{-1}))$ where k is the Boltzmann constant in $eV K^{-1}$ and $\text{logit}(x) = \ln(x/(1 - x))$. Model comparison between the monotonic Arrhenius and unimodal Sharpe–Schoolfield descriptions of respiration used approximate leave-one-out cross-validation by Pareto-smoothed importance sampling¹⁹. Unless stated, reported values are posterior medians with 95% credible intervals.

Model-free optimum of the apparent CUE. Writing $G = r \cdot q \cdot N$ and $R = c \cdot K$, with N the cell density at the start of the fit window, $CUE = 1/(1 + (c/q) \cdot K/(r \cdot N))$; the temperature that maximises CUE is the temperature that minimises $K/(r \cdot N)$, and c and q cancel. With a common inoculum and growth at the fitted rate over the interval t_s from inoculation to the window start, $N = N_{inoc} \cdot e^{rt_s}$ and the quantity minimised is $y = \ln K - \ln r - r \cdot t_s$, in which N_{inoc} cancels. The estimate takes the mean of y over growing series at each measured temperature, locates the grid minimum, refines it with a three-point parabola through the minimum and its neighbours, and obtains an interval and $P(T_{opt} < 37 \text{ }^\circ\text{C})$ by resampling isolates within group and then wells within isolate and temperature (2,000 draws). The earlier form of this estimate omitted the e^{rt_s} term and resampled wells within temperature only; both versions are given in `results/tables/modelfree_topt_n0.csv` (`scripts/75_modelfree_topt_n0.py`). The activation energies of the fitted r and of the volumetric K were compared on the same basis.

Per-well likelihood model for growth. Whether a well grew was decided for every series by comparing two nested models of the oxygen trace after the 1.5 h equilibration exclusion, over at most 15 h, decimated uniformly to 300 points. Both have a free plateau F , the level at which the trace flattens whatever its absolute value: a linear decline to the plateau, $O_2(t) = \max(O_{2,0} - Kt, F)$, and an exponential decline to the plateau, $O_2(t) = \max(O_{2,0} - (K/r)(e^{rt} - 1), F)$ with $r \geq 0$, the linear model being the exponential one at $r = 0$. The maximum is smoothed over 0.15 mg L^{-1} . Each mean was fitted by nonlinear least squares, the residuals were described by a Gaussian AR(1) process with its autocorrelation and innovation variance profiled from them, and the exact AR(1) log-likelihood was evaluated for each model. Evidence for growth is the likelihood ratio $LR = 2(\ln_{\text{exp}} - \ln_{\text{lin}})$. Because $r = 0$ lies on the boundary of the parameter space the χ^2 approximation does not apply, so the null distribution of LR was obtained for every well by parametric bootstrap from its own fitted linear model with AR(1) noise (499 draws); the 95th percentile of that null is typically 5 to 10 rather than 3.84. A well is scored as growing when the bootstrap p-value is below 0.01, as respiring without detectable growth otherwise, and as showing no detectable respiration when the total drawdown is below 1.0 mg L^{-1} . The model returns r , K and F for every well; the fitted r in Fig. 1b is this quantity. Growth wells and non-growth wells are cleanly separated on this statistic (median LR 0 and 90th percentile 2.3 among non-growth wells; growth wells begin at LR 11).

High-oxygen refit. For every well in the primary fits, the oxygen-dynamics model was refitted on the manual window truncated at a fixed cutoff of 50% of the measured oxygen at the window start, reproducing the pipeline’s full-window fit as a check (median ratio to the pipeline’s r and K , 1.00 and 0.99), and the model-free CUE optimum and the Arrhenius activation energy of K were recomputed from both arms with the same isolate-first resampling. The per-well likelihood model was also rerun on traces truncated at the same cutoff. Scripts and outputs in `reprocess/hi02`; Supplementary Note 4.

Optical-density test of the growth calls. Because a trace $K \cdot e^{rt}$ from a growing population is identical to one from a constant population whose per-cell respiration rises at rate r , the growth calls were tested against an independent measure of cell mass rather than by simulation. Four isolates were grown in Greiner 24-well plates (1 mL YMS per well, five wells per isolate, three medium-only blanks, lid on) at 37, 40 and 42 °C in a Tecan Infinite 200Pro, one plate per temperature on consecutive days, from fresh overnight YMS pre-cultures at 30 °C diluted to $OD_{600} 0.05$ (cuvette). OD_{600} was read every 20 min for 16.3 h with 20 s of 6 mm orbital shaking before each read and no movement between reads. Blank-corrected OD was compared with the likelihood-model calls for the same isolate in the sealed PreSens vials at the same temperature (the 38 °C vials for the 37 °C plate, since the vials were run on a 2 °C grid). The plate is aerobic and shaken while the vials are sealed and static, so the comparison is of whether growth occurred and of the ordering of isolates, not of absolute rates. Data in `data/od_validation`; script `scripts/81_od_validation.py`; Supplementary Fig. 13.

Sensitivity arm: the pre-registered classifier. Before the likelihood model was written, the state of every series had been derived by a three-criterion classifier applied blind to taxon, isolate and temperature, with its protocol, source and output hashed before any label was restored (protocol versions 1 to 4, archived with their hashes). It scored a well as confirmed growth when all of blocked cross-validation, a bootstrap floor on r and $rt \geq 0.5$ agreed, and as ambiguous when they disagreed. The likelihood model agrees with it on 930 of its 931 confirmed-growth wells. Of its 154 ambiguous wells, 90 failed only the cross-validation criterion because the fixed fit window ran into the plateau, and 89 of those are growth under the likelihood model; of the remaining 64, 13 are growth. Of its 114 respiration-only wells, 31 show growth under the likelihood model (LR 11 to 220, median $r 0.09 \text{ h}^{-1}$), curvature the classifier missed because its window extended 15 h into the plateau. Every count in Figs 1 to 3 uses the likelihood model; the classifier’s states are retained in the project archive.

Censoring and the tests applied to the limits. The temperature of growth loss for an isolate is the lowest measured temperature at which fewer than half of its wells grow, with that loss persisting at every higher measured temperature; isolates still growing at 44 °C are right-censored as $>44 \text{ °C}$ and never imputed to a numerical value. Loss temperatures were compared between *C. auris* and the relatives by the log-rank statistic with an exact permutation null (20,000 permutations). Proportions at a given temperature are reported with isolate-clustered 95% intervals, since the five wells of an isolate at one temperature are not independent replicates of a species.

Interval-censored model of the limit. Each isolate’s loss temperature was taken to lie in $(T_{\text{last}}, T_{\text{loss}}]$, the 2 °C grid step over which growth was lost, or in $(44\text{ °C}, \infty)$ for isolates still growing at the ceiling. Two parametric forms were fitted by maximum likelihood with a *C. auris* indicator: a normal location-shift model, whose coefficient is the shift in the limit in °C, and a Weibull proportional-hazards model on temperature above 30 °C, whose coefficient gives the hazard ratio of growth loss. Confidence intervals are profile-likelihood intervals and tests are likelihood-ratio tests; both were refitted without Hae_1724 (shift +4.8 °C, 3.2 to 6.8). Script `scripts/66_survival_lik.py`.

Isolate-level crossings and their concordance with the limit. The cost-multiplier crossings were recomputed for every isolate from its own curve in the hierarchical fit (group effect plus that isolate’s random effects on the growth activation energy, the deactivation midpoint and the respiration activation energy), for every posterior draw as for the group-level analysis. Each isolate’s 3× crossing (posterior median) was compared with its observed loss of growth under the likelihood model: the signed distance to the grid interval (zero if inside), the posterior probability that the crossing falls inside the interval, Spearman’s rank correlation across the 20 isolates with censored limits tied above 44 °C, and an interval-censored regression of the limit on the crossing. Scripts `scripts/65_isolate_crossings.R` and `scripts/67_safety_margin_lik.py`. Whether any fitted quantity predicts the limit better than another was tested by using each in turn as the single predictor in the interval-censored model (the 1.5×, 2× and 3× crossings, the growth optimum, the CUE optimum, the deactivation midpoint T_h and the temperature at which fitted growth falls to half its peak), compared by maximised log-likelihood and leave-one-isolate-out predictive density; the half-peak temperature and the crossings were indistinguishable and both outperformed T_h and the CUE optimum (`scripts/69_limit_predictors_lik.py`).

Decomposition of the displacement. For each of the 12 *C. auris*-clade-versus-relative pairs and 600 posterior draws of the group-level parameters, the 3× crossing was recomputed with every subset of the four curve parameters (E , E_h , T_h , E_R) taken from the *C. auris* clade and the rest from the relative; each parameter’s contribution is its Shapley value over those subsets, so the four contributions sum to the displacement in every draw. Shares are medians over draws, averaged over pairs. E_h is estimated at group level only and is weakly identified for *C. duobushaemulonii*, whose growth is lost across most of the upper grid, so its contribution in the three *C. duobushaemulonii* pairs is the least certain. Script `scripts/70_displacement_decomposition.py`; per-pair values in `results/tables/displacement_decomposition.csv`.

Respiration across the limit. The initial oxygen consumption rate K of every well with detectable respiration (1,199 wells) was regressed on inverse temperature with isolate fixed effects, an indicator for wells at or above the isolate’s own growth limit, and, in a second model, an interaction of that indicator with inverse temperature; standard errors were clustered by isolate and temperature. The indicator’s coefficient is the log offset of above-limit wells from the isolate’s sub-limit Arrhenius line. Censored isolates contribute only sub-limit wells. The bound on how much of the across-limit fall in K is cell number rather than per-cell rate uses the growth accrued before the window: for each isolate with a limit inside the assay, the median e^{rt_s} over its wells two degrees below the limit (r the likelihood-model rate, $t_s = 1.5\text{ h}$), by which the sub-limit K is divided before the ratio of the median K two degrees above the limit to that two degrees below is recomputed; per-isolate factors, raw and bounded ratios are in `results/tables/prewindow_growth_factor.csv` (medians 1.16, 0.26 and 0.28 over the nine isolates with a measured temperature two degrees above their limit). Script `scripts/71_respiration_across_limit.py`.

Kinetics within non-growing wells. Each well respiring without detectable growth was refitted with a consumption rate decaying exponentially from its initial value, $O_2(t) = O_{2,0} - (K/)(1 - e^{-t})$ with the same plateau term, against the constant-rate model, by the AR(1) likelihood ratio used for the growth call; a decline was accepted at $LR > 10$. Among declining wells the decay rate was regressed on inverse temperature with group intercepts and isolate-clustered errors ($E = -1.2 \pm 0.7\text{ eV}$, $p = 0.10$). Script `scripts/72_arrest_kinetics.py`.

Fever profiles. Daily growth under a temperature profile was computed per isolate and posterior draw as $r(T(t))dt$ over 24 h from the isolate’s fitted growth curve, with r in h^{-1} obtained from the per-cell flux and the group carbon quota as in Fig. 4, assuming an instantaneous thermal response and no mortality; profiles

were constant 37, 38.5 and 40 °C, 12 h at 40 °C with 12 h at 37 °C, and 4 h at 40 °C with 20 h at 37 °C. Script `scripts/73_fever_profiles.py`.

Isolate-level trait correlations. For each isolate and posterior draw the peak growth rate (h^{-1} , from the per-cell flux and the group quota), the growth optimum, the CUE optimum and the 37 to 40 °C change in respiration per unit growth were computed from the isolate’s own curve, and the growth limit was taken from the likelihood model with censored limits imputed uniformly on (44, 48]. Spearman’s ρ between the limit and each trait was computed in every draw, over all 20 isolates, within *C. auris* and within the relatives; posterior medians and 95% CrI are reported. Script `scripts/74_tradeoffs.py`.

Phylogenetic signal in the limit. The isolate-level loss temperature was modelled as $limit_i = \mu + u_{g(i)} + e_i$, with u a multivariate normal group effect whose covariance is the shared root-to-ancestor path length between the seven groups on the supermatrix tree (re-rooted at their common ancestor and scaled to unit mean variance), and e an independent isolate residual. The share $h^2 = \sigma_p^2 / (\sigma_p^2 + \sigma_e^2)$ was fitted by maximum likelihood on a grid; with isolates attached at their group’s tip it is Pagel’s λ . Censored isolates (>44 °C) were imputed uniformly on (44, 48] over 200 imputations and the estimates pooled. Significance was assessed by a likelihood-ratio test of $\sigma_p^2 = 0$ against the boundary σ_p^2 mixture and by a permutation null shuffling the limits across isolates (20,000 draws). Setting the censored limits to 44 or 46 °C instead gives $h^2 = 0.68$ and 0.58. Script `scripts/64_phylo_signal_lik.py`.

Oxygen consumption after growth loss. Consumption was compared on a fixed, prespecified window (90–390 min: the 1.5 h equilibration exclusion plus the 5 h early window used throughout), applied identically to every well. Traces reaching the oxygen floor inside that window were treated as right-censored rather than as low metabolism; because growing wells reach the floor far more often than non-growing wells, the reported ratio of non-growing to growing consumption is an upper bound rather than a point estimate. The share of consumption in growing wells is the oxygen consumed on that window by an isolate’s growing wells divided by that consumed by all its wells at the same temperature.

Predictive test. Whether sub-febrile metabolism forecasts high-temperature failure was tested using only fits restricted to 22–38 °C, validated by leave-one-isolate-out with the complete isolate withheld, and scored by mean absolute error against the observed temperature of growth loss (likelihood-model limits, censored isolates scored as correct when predicted ≥ 44 °C) and by held-out log predictive score for the binary outcomes. The concordance between the cost-multiplier crossings and the observed loss temperature was computed per taxon from a logistic fit of the well-level growth calls against temperature, over the taxa whose loss lies inside the assay range.

Phylogenomics and proteome comparison. Ten annotated proteomes were used (NCBI accessions GCF_002759435.1, GCF_003013715.1, GCF_002775015.1 and GCA_008275145.1 for *C. auris* clades I–IV; GCF_002926055.2, *C. haemulonii*; GCF_002926085.2, *C. duobushaemulonii*; GCF_003013735.1, *C. pseudohaemulonii*; GCF_000182765.1, *C. parapsilosis*; GCF_000182965.3, *C. albicans*; GCF_014636115.1, *C. lusitaniae*). Reciprocal best hits were computed with DIAMOND blastp²⁰ at $e < 10^{-10}$, requiring 1:1 orthology in every genome, giving 2,931 single-copy orthologs; a random sample of 600 was aligned with MAFFT²¹, columns with more than 20% gaps were trimmed, and the concatenated supermatrix (289,787 aa $\times$ 10 taxa) was used to infer a maximum-likelihood tree with FastTree 2 under the LG model with CAT rate categories and Gamma20 likelihood rescaling (`-lg -gamma`)²², rooted on *C. albicans*, *C. parapsilosis* and *C. lusitaniae*. Every node has SH-like support of 1.00. The tree was rebuilt from the same proteomes with recorded versions (DIAMOND 2.1.9, MAFFT 7.505, FastTree 2.1.11; `phylo/trees/rerun_2026-09-29`): the more sensitive DIAMOND release recovers 3,658 single-copy orthologs, and a 600-gene sample from them gives the identical topology with every node at 1.00 and patristic distances within 5% of the reference tree used here. Whole-proteome amino-acid identity between clades was taken from the same reciprocal-best-hit alignments.

Transcriptional data. The clade-resolved comparison used a public RNA-seq dataset comparing two *C. auris* clades under one controlled condition (GEO GSE165762)²³, in which three clade I clinical isolates and the clade II type strain CBS10913 were grown in YPD at 37 °C for 4 h. Central-metabolic enzyme genes were compared against the genome background by a two-sided Wilcoxon rank-sum test on $\log_2$ fold-change.

Reporting summary. Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Software and reproducibility. Analyses were run in R (brms, posterior, loo, minpack.lm, ggplot2, patchwork) and Python (NumPy, SciPy, pandas, Matplotlib); the exact environments are pinned in the project's `renv.lock` and `requirements.txt`. Every figure is produced by a named script, listed in the repository README, and the per-well likelihood model (`reprocess/lik_model.py`), the pre-registered classifier protocol, its hashes and the sealed blinding key are retained in the project archive.

Data availability

The raw PreSens oxygen exports for all 1,200 wells, the manually selected fit window for every trace, the blinded trace archive with its sealed key, the per-well likelihood-model results, the posterior draws of every fitted curve and the source data behind every figure are deposited at [REPOSITORY: Zenodo DOI to be inserted before submission]. The public RNA-seq dataset re-analysed here is GEO GSE165762, and the genome assemblies are listed under Phylogenomics in Methods.

Code availability

All analysis code, including the oxygen-dynamics fitting, the per-well likelihood model (`reprocess/lik_model.py`), the pre-registered classifier with its four hashed protocol versions, the hierarchical Stan models and the script producing each figure, is at [REPOSITORY: GitHub URL and archived Zenodo DOI to be inserted before submission], with the environments pinned in `renv.lock` and `requirements.txt`. The README maps every figure to the script that draws it.

Acknowledgements

[To be added.]

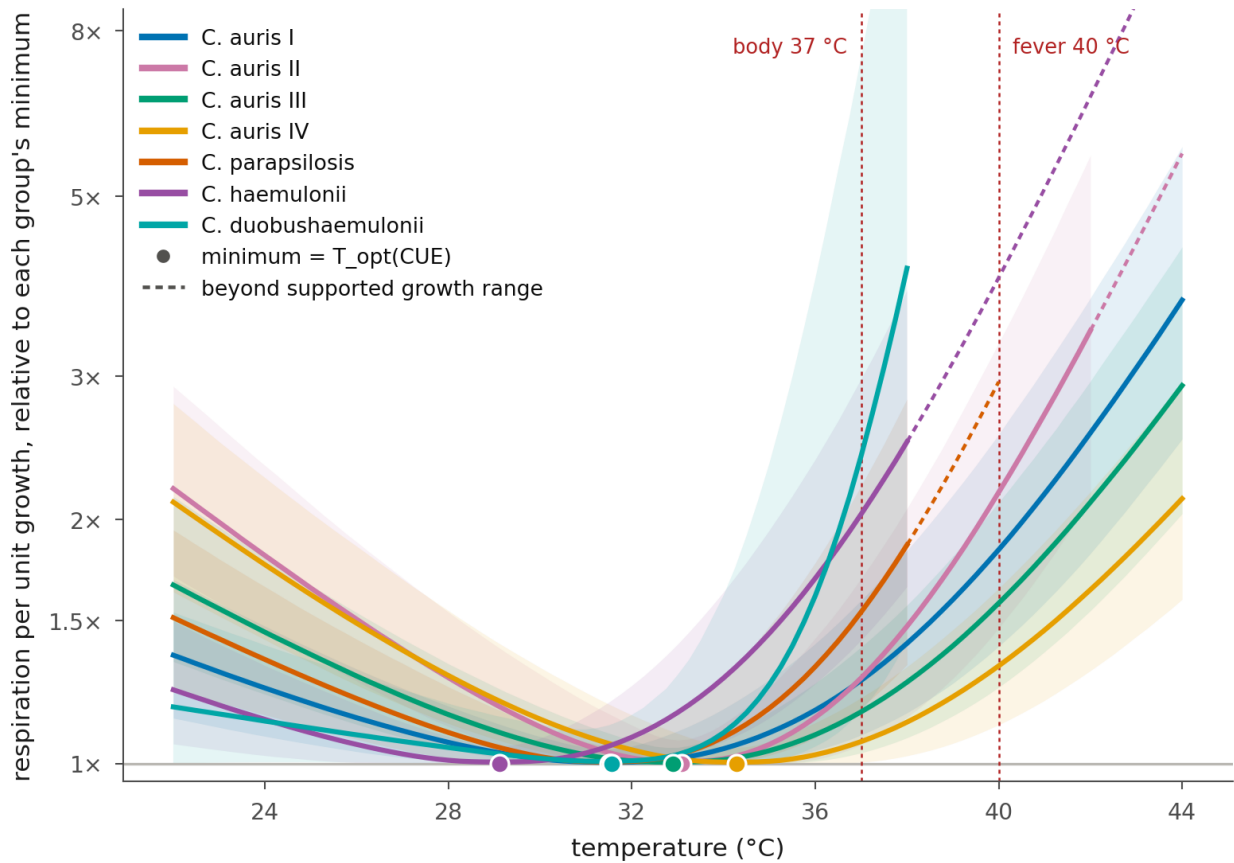
Author contributions

I.C. designed the study, performed the respirometry and the optical-density validation, inspected and windowed every trace, wrote the analysis code, analysed the data and wrote the manuscript. N.A.R.G. provided the isolates and advised on *Candida* biology. N.S. advised on the metabolic interpretation. S.P. contributed to the thermal performance modelling. G.Y.-D. conceived and supervised the study. All authors commented on the manuscript. [Draft: co-authors to confirm.]

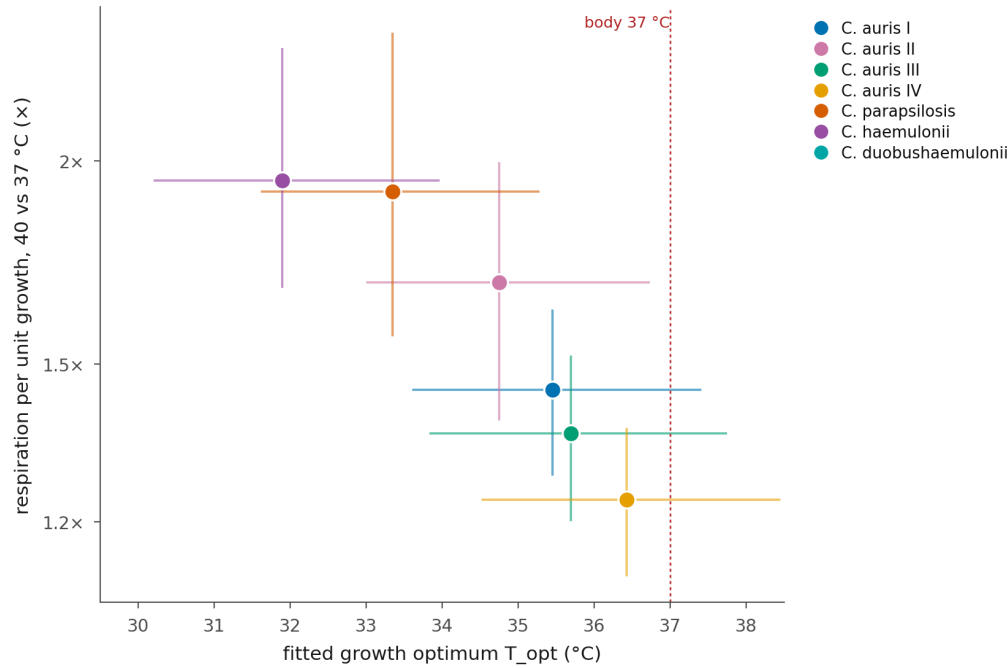
Competing interests

The authors declare no competing interests.

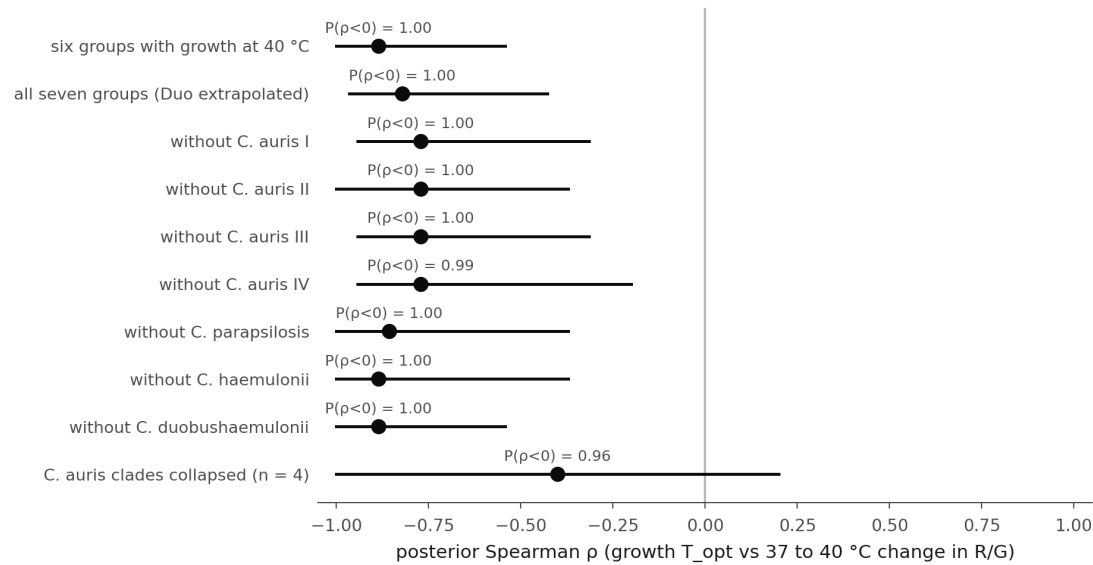
Supplementary figures



Supplementary Figure 1. Temperature dependence of the respiration-to-growth ratio. Respiration per unit growth at each temperature, divided by that group's own minimum, so that 1× marks each group's cheapest temperature and the filled point marks the position of that minimum, $T_{\text{opt}}(\text{CUE})$ (the quantity plotted in Fig. 5b). Bands are 95% credible intervals; dashed lines mark body (37 °C) and fever (40 °C) temperature. Curves are solid over the range in which at least half of the group's wells grew and dashed beyond it.

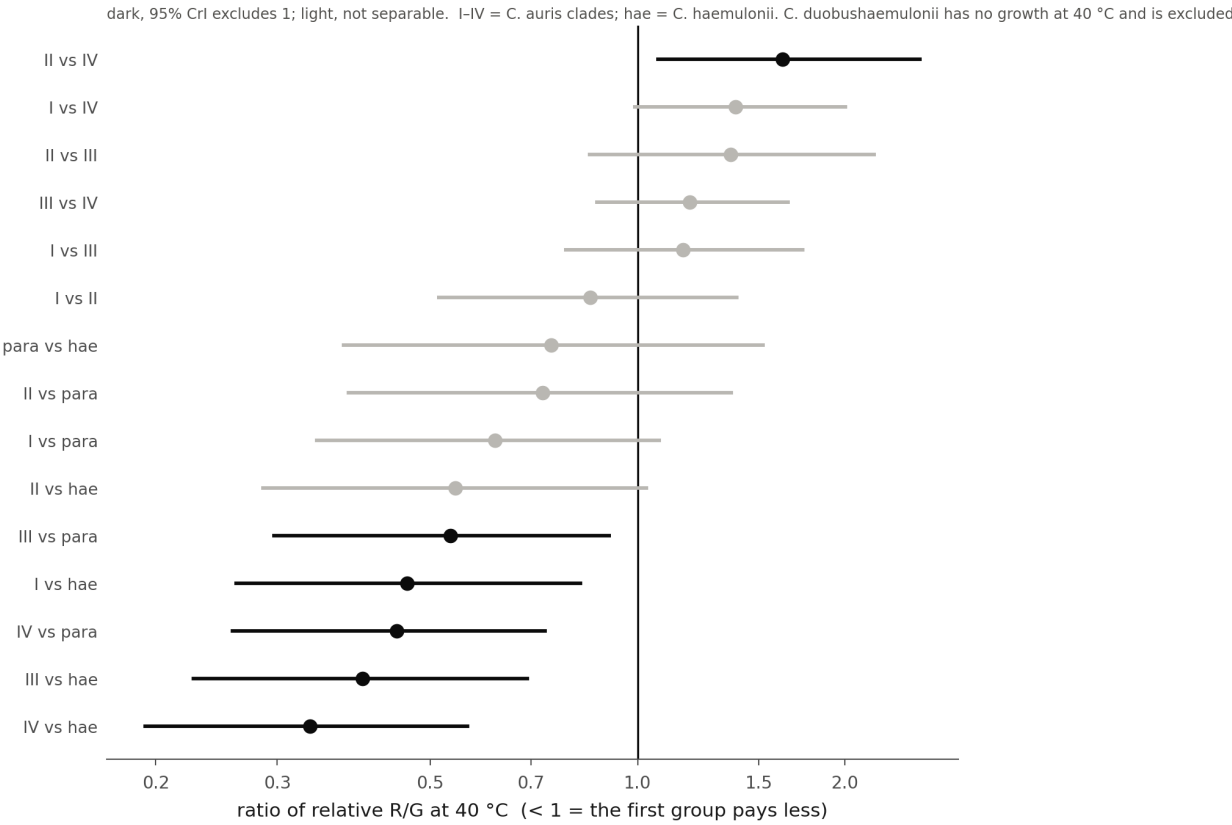


Supplementary Figure 2. Growth optimum against the cost of a fever, across the six groups with measurable growth at 40 °C (*C. duobushaemulonii* excluded). Horizontal bars, 95% CrI on the fitted growth optimum; vertical bars, 95% CrI on the 37 to 40 °C change in respiration per unit growth. Both axes are derived from the same fitted curves and four of the seven points are clades of one species, so the association is reported as exploratory. The *C. duobushaemulonii* cost extrapolates the fitted curve beyond 38 °C, where that group has almost no measurable growth, and is not quoted in the text.

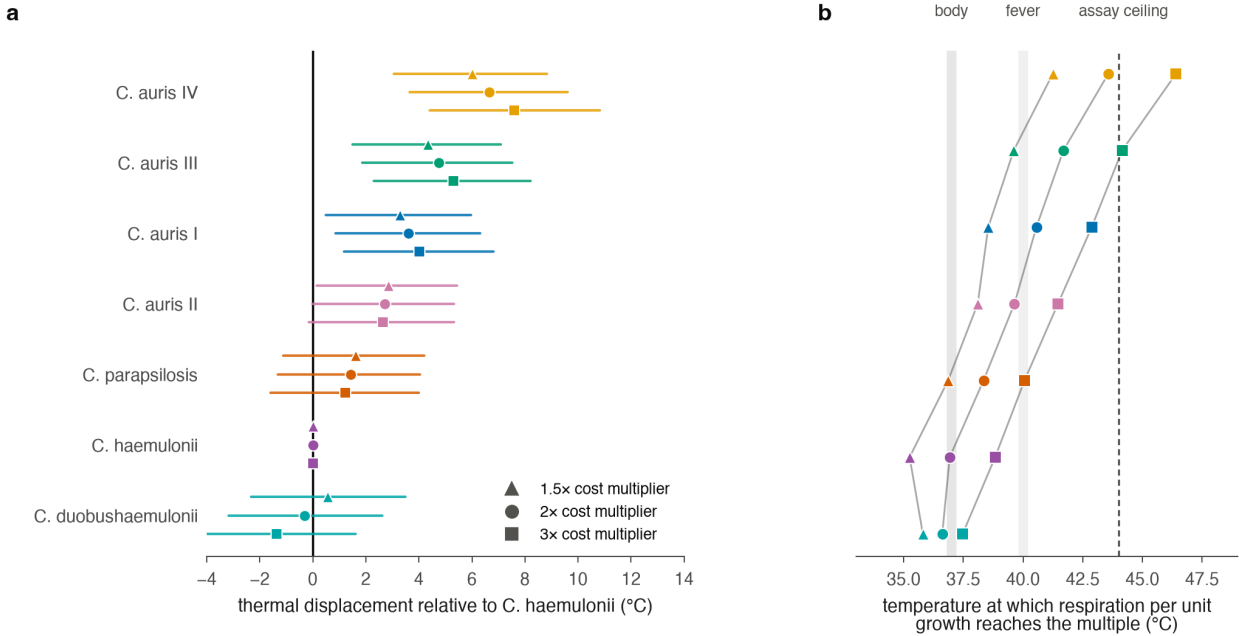


Supplementary Figure 3. Leave-one-out robustness of that association. Posterior Spearman ρ between the growth optimum and the 37 to 40 °C change in respiration per unit growth, computed over the six groups with measurable growth at 40 °C, over all seven with *C. duobushaemulonii*'s extrapolated cost included, then dropping each group in turn, and finally with the four *C. auris* clades averaged into a single *C. auris* value in every posterior draw before the correlation is taken, so that the four clades of one species count once rather than four times (four taxa in that row). Points are posterior medians, bars 95% CrI, with the posterior

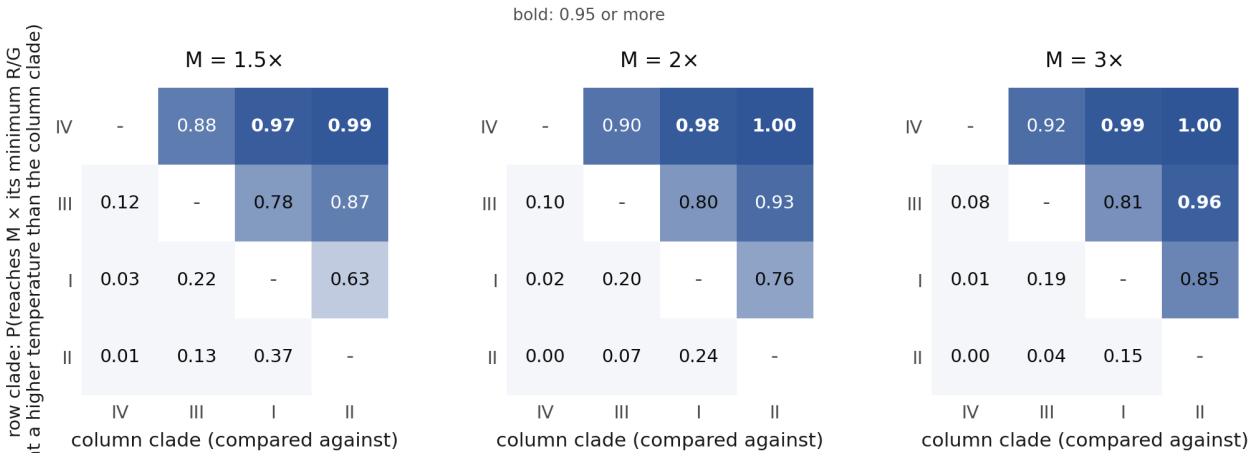
probability that $\rho < 0$ annotated.



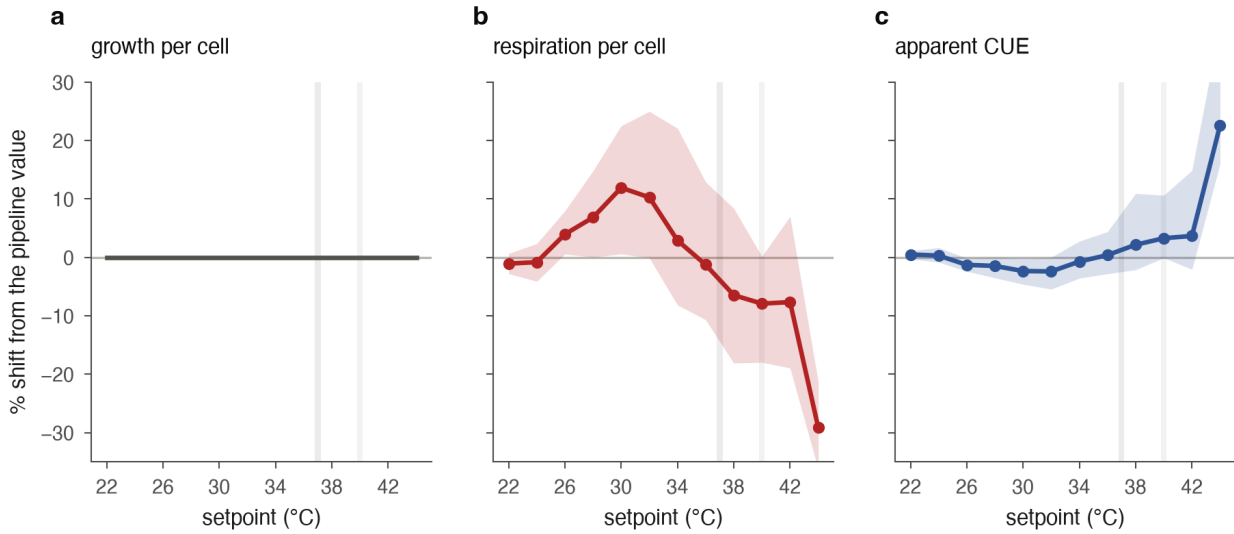
Supplementary Figure 4. Posterior pairwise contrasts in the relative respiration-to-growth ratio at 40 °C, for the 15 pairs among the six groups with measurable growth at 40 °C (*C. duobushaemulonii* excluded). Ratios below 1 mean the first group pays less; dark bars mark the six contrasts whose 95% CrI excludes 1, light bars the nine that do not separate.



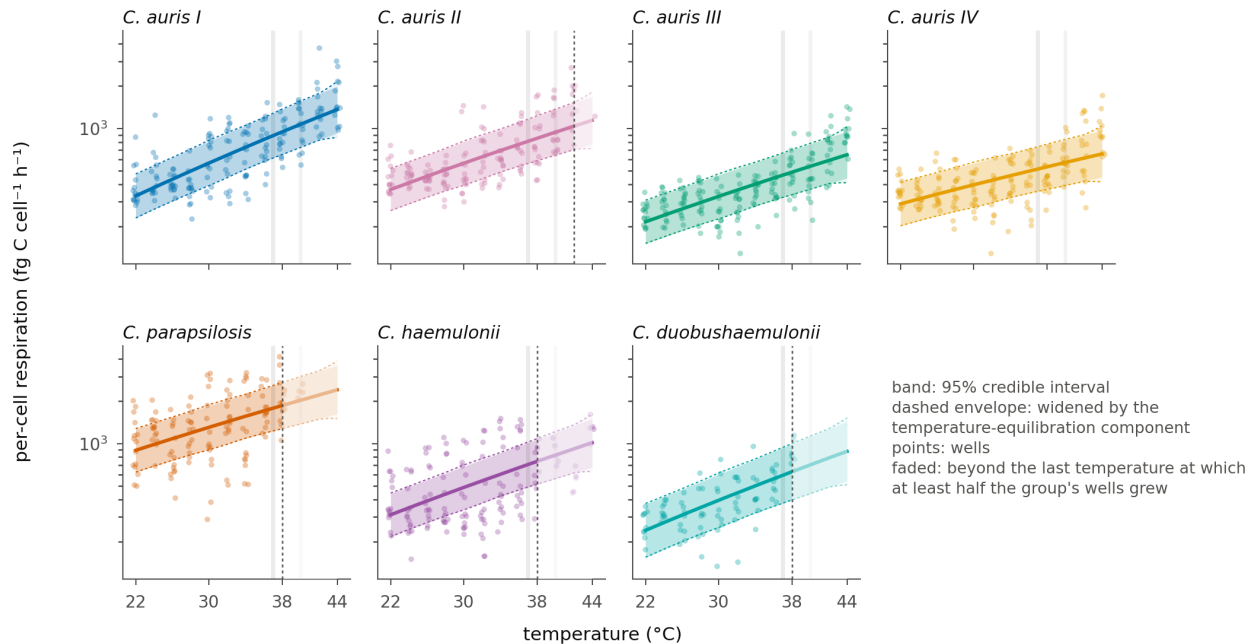
Supplementary Figure 5. *C. auris* is displaced along the temperature axis. Derived from the manual fit windows. (a) Displacement relative to *C. haemulonii* at three cost multipliers (posterior median, 95% CrI). (b) Absolute crossing temperatures; shaded bands, 37 and 40 °C; dashed line, the 44 °C assay ceiling. The rank order is identical at every multiplier. No multiplier is a biological failure threshold; 3× lies closest to observed growth loss (mean absolute error 0.73 °C against 2.29 °C for 2×, over the four taxa whose loss lies inside the assay range). The sign of the margin relative to host temperature is multiplier-dependent and is not used as a criterion. Above 38 °C the *C. haemulonii* estimate is strongly informed by one isolate, Hae_1724.



Supplementary Figure 6. Pairwise ordering of the *C. auris* clades on the cost-multiplier axis does not depend on the multiplier. Each cell is the draw-wise posterior probability that the row clade reaches M times its own minimum respiration per unit growth at a higher temperature than the column clade, at M = 1.5, 2 and 3. Only the two extremes separate decisively (Clade IV versus II, $P \geq 0.99$ at every M); adjacent clades do not, so the clades are reported as a gradient rather than a ranking. The crossing temperatures themselves are shown in Supplementary Fig. 5b.

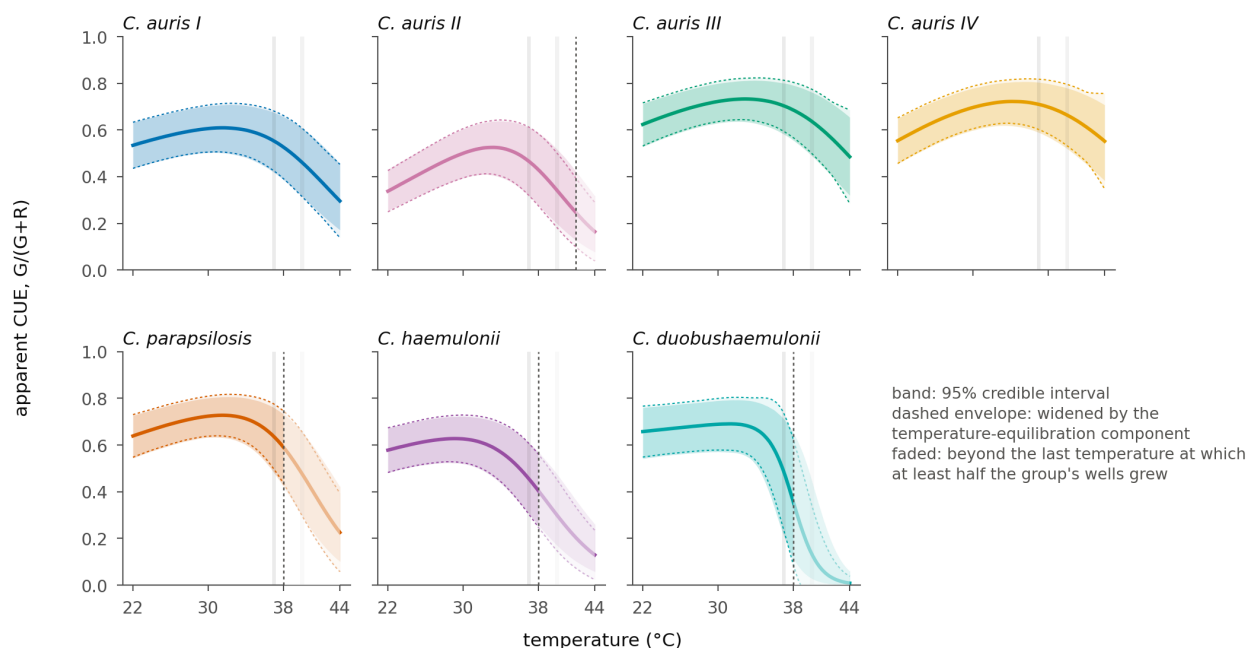


Supplementary Figure 7. Sensitivity of each per-cell quantity to temperature equilibration. Median percentage shift from the pipeline value versus setpoint, estimated from the logged internal vial temperature against each group's growth curve (band, interquartile range across the vials at that setpoint, so it spans the between-vial spread of the correction rather than posterior uncertainty; the correction is propagated by recomputing N_0 from the logged temperature and each group's own growth curve and re-deriving every per-cell quantity from that N_0), for (a) growth per cell, (b) per-cell respiration and (c) the apparent CUE. Growth per cell is immune because it does not use N_0 ; per-cell respiration shifts by a few percent through the mid-range and up to about -27% at 44°C ; CUE follows, buffered.

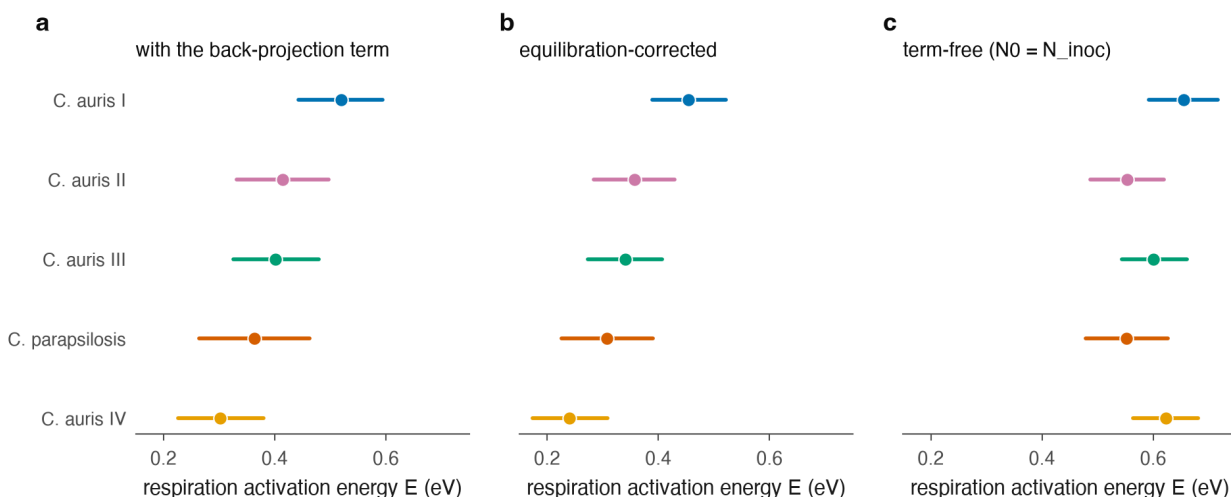


Supplementary Figure 8. Per-cell respiration versus temperature (Bayesian Arrhenius, log scale), per group, with the temperature-equilibration uncertainty overlaid. Points, wells; band, 95% credible interval; dashed envelope, the same interval widened by the equilibration component; the faded region beyond the dotted line marks temperatures at which fewer than half the group's wells grew, where the curve is a fitted extension and not an observation. The two intervals coincide through the mid-range and separate only

toward the extremes; respiration's monotonic rise over the observed range is unaffected.

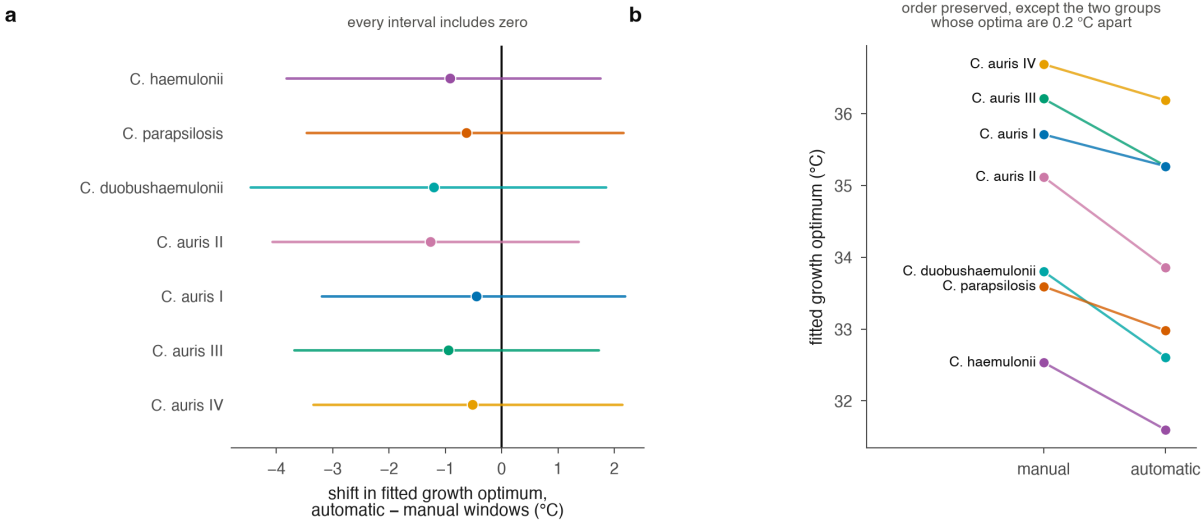


Supplementary Figure 9. Apparent carbon-use efficiency versus temperature, per group, with the temperature-equilibration uncertainty overlaid. Solid band, 95% credible interval; dashed envelope, widened by the equilibration component. The two coincide through the mid-range and separate toward the extremes. Under this correction the fitted optimum of the warmest clade approaches body temperature (*C. auris* Clade IV, 36.6 °C, $P(< 37\text{ °C}) = 0.80$); this is the weakest of the three treatments for that clade and is reported alongside them in the main text.

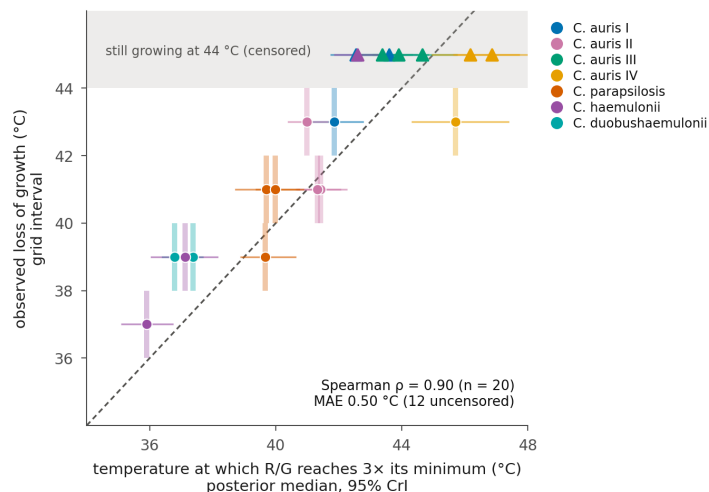


Supplementary Figure 10. The between-group respiration activation energy under three treatments of the inoculum back-projection, for the five groups of the original five-group fit. Each panel shows the hierarchical Bayesian Arrhenius activation energy E_R per group (posterior median; bars, 95% credible interval), refitting the same model under: (left) the published back-projection, $N_0 = N_{\text{inoc}} \cdot \exp(r \cdot \Delta t)$; (centre) the equilibration-corrected N_0 , recomputed from the logged internal vial temperature over the onset window; and (right) the term removed entirely, $N_0 = N_{\text{inoc}}$. The *C. auris*-versus-*C. parapsilosis* separation is preserved across all three, but the finer ordering within *C. auris* is not: Clade IV holds the lowest activation

energy under the published and equilibration-corrected treatments and moves to fourth of five when the term is removed. The term-free case is not the physical null (the inoculum grows before the fit window), so it bounds the correction from the opposite side rather than representing the correct value; it is shown to make the dependence explicit.

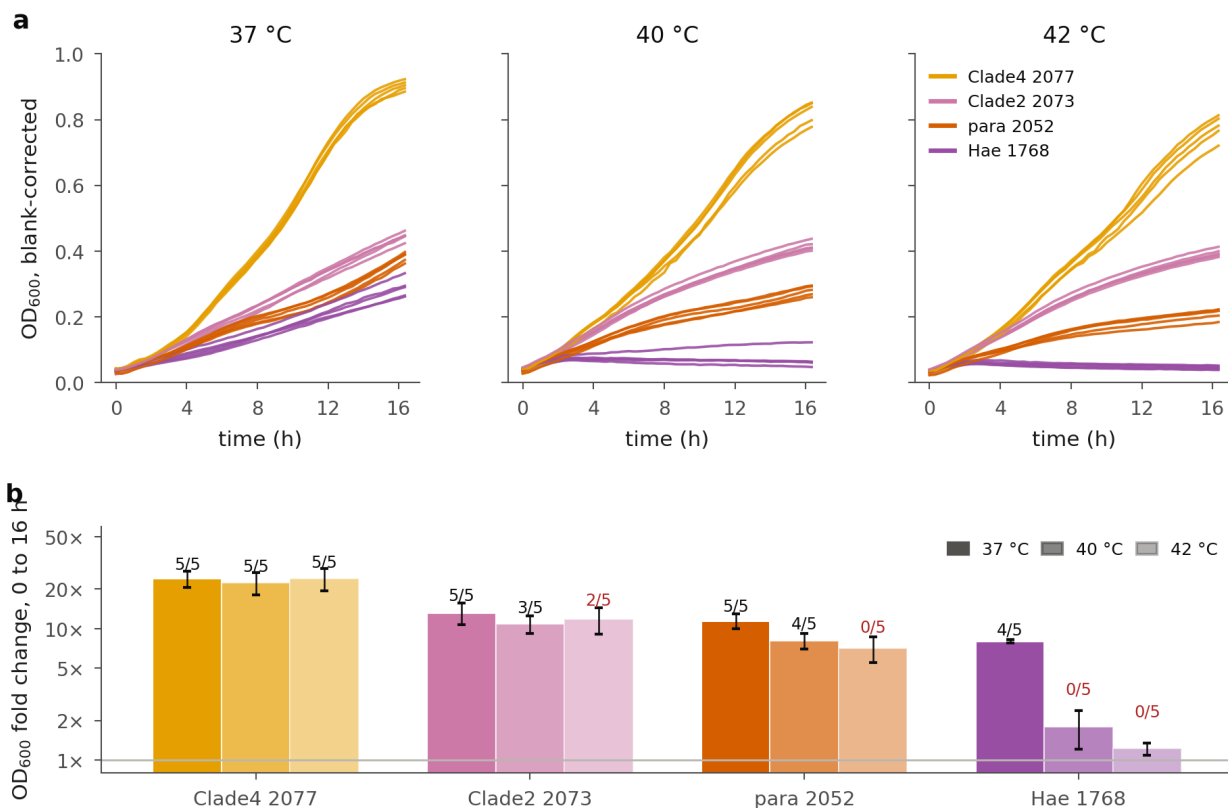


Supplementary Figure 11. Manual against automatic choice of fit window. The same hierarchical growth model was refitted on the same traces under both rules, the manual windows of the analysis pipeline and the fixed rule of the pre-registered classifier (window from 90 min to 98% oxygen depletion), so the two arms differ only in how each window was chosen. **(a)** Shift in the fitted growth optimum, automatic minus manual, with 95% CrI; every group shifts down by 0.45 to 1.26 °C and every interval includes zero (posterior probability of a positive shift 0.17 to 0.37). **(b)** The optima themselves. The order is preserved except for *C. parapsilosis* and *C. duobushaemulonii*, whose optima differ by 0.2 °C under the manual rule and 0.4 °C under the automatic one and which exchange places. Activation energies move by at most 0.19 eV and the deactivation midpoint by at most 0.83 °C, except in *C. auris* II (2.82 °C), the group that loses most data under the automatic rule. The manual arm of this refit reproduces the published optima with a systematic offset of +0.24 to +0.64 °C, traced to prior differences between the refit script and the analysis pipeline (a narrower prior on E_h and unbounded T_h); because both arms here use identical priors, that offset cancels in the manual-minus-automatic contrast reported above. The primary fits were not re-run under the refit script's priors because the comparison needed here is between window rules at fixed priors, which this design gives directly; the primary priors are the ones reported in Methods and the offset is reported so that the two sets of optima are not read as independent.



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797 **Supplementary Figure 12.** The 3× crossing of the respiration-to-growth ratio tracks the growth limit
 798 isolate by isolate. Each point is one isolate: the temperature at which its respiration per unit growth reaches
 799 three times its own minimum (posterior median and 95% CrI from the isolate's own curve in the hierarchical
 800 fit) against the temperature at which growth is lost under the per-well likelihood model (vertical bar, the 2
 801 °C grid interval; triangles, isolates still growing at 44 °C, plotted in the shaded band). Blue, *C. auris*; orange,
 802 relatives; dashed line, identity. Spearman $\rho = 0.90$ over 20 isolates; mean absolute error 0.50 °C over the
 803 twelve uncensored isolates; at 1.5× and 2× every crossing lies below the observed interval (mean error −3.0
 804 and −1.5 °C). An interval-censored regression gives slope 0.87 and rejects the identity line ($p = 0.006$), the
 805 compression coming from the assay ceiling at the warm end.



806

807 **Supplementary Figure 13.** Independent test of the oxygen-derived growth calls by optical density.

Four isolates spanning the states the likelihood model assigns (Clade4_2077, growth at every temperature; Clade2_2073, mixed at 40 to 42 °C with non-growing vials respiring at the rate of growing ones; para_2052, growth lost at 42 °C; Hae_1768, growth lost at 40 °C) were grown in 24-well plates (1 mL YMS, five wells each, OD₆₀₀ 0.05 at inoculation) at 37, 40 and 42 °C in a temperature-controlled plate reader, one plate per temperature. The plates are aerobic and shaken before each read, whereas the PreSens vials are sealed and static, and the 37 °C plate is compared with the 38 °C vials because the vials were run on a 2 °C grid; the comparison is therefore of whether growth occurred and of the ordering of isolates, not of absolute rates. (a) Blank-corrected OD₆₀₀ read every 20 min for 16 h; every line is one well. (b) Fold change in OD over the run (mean ± SD of five wells) at 37, 40 and 42 °C (dark to light), labelled with the number of the same isolate's sealed PreSens vials (of five) in which the likelihood model detected growth at that temperature (38 °C vials for the 37 °C plate); red where fewer than three. Where the model detected growth, OD rose 7- to 24-fold; where it detected none in Hae_1768, OD rose only 1.2- to 1.8-fold over 16 h, a single division in the first 2.5 h followed by no further increase, beside wells of Clade4_2077 that rose 20-fold. The rank order of the four isolates is the same by both methods at every temperature. In the two isolate-temperature combinations at the edge of the vial limit (para_2052 and Clade2_2073 at 42 °C) the aerobic, shaken plate supported slow growth (7- and 12-fold over 16 h, most of it in the first 3 h) that the sealed, static vial did not register, so “no detectable growth” in a vial is a conservative call. Absolute rates are not compared because the plate becomes oxygen-transfer limited within hours, which is visible as the bend in every curve.

Supplementary notes

Supplementary Note 1 | Predictive test. Predictors were computed from the manual fits restricted to 22 to 38 °C only: the fitted growth rate at 38 °C, the group growth optimum, log(K/r) at 38 °C and its slope over 34 to 38 °C. Three models were compared: growth predictors only (M1), metabolic predictors only (M2) and both (M3). Continuous outcome: the temperature of growth loss under the likelihood model, scored by mean absolute error under leave-one-isolate-out with the complete isolate withheld, over the twelve isolates whose loss lies inside the assay range (M1 0.93 °C, M2 1.42 °C, M3 1.10 °C). Binary outcomes: growth at 40, 42 and 44 °C and respiration without growth at 40 °C, scored by held-out log predictive density (M1 -0.20, -0.26, -0.46, -0.20; M2 -0.47, -0.47, -0.53, -0.47; M3 -0.38, -0.29, -0.46, -0.38) and Brier score (M1 0.056, 0.078, 0.130, 0.056; M2 0.144, 0.153, 0.174, 0.144; M3 0.127, 0.088, 0.141, 0.127). The growth-only model scored best on every outcome (results/tables/phase3_loio_results_lik.csv).

Supplementary Note 2 | Concordance between the cost-multiplier crossings and the growth limit. Of the three multipliers, 3× lies closest to the temperature at which growth is lost (mean absolute error 0.73 °C over the four taxa whose loss lies inside the assay range, against 2.29 °C for 2×; Methods). The same holds isolate by isolate: computing each isolate's own crossings from its curve in the hierarchical fit, the 3× crossing tracks that isolate's observed loss of growth with a rank correlation of 0.90 across all 20 isolates and a mean absolute error of 0.50 °C over the twelve whose loss lies inside the assay range (Supplementary Fig. 12). Because the same fitted growth rate enters both quantities this is a concordance rather than an independent validation, and the crossing carries no information about the limit beyond the growth curve itself: the temperature at which each isolate's fitted growth rate falls to half its peak coincides with its 3× crossing (r = 0.99 across isolates, mean difference 0.2 °C) and predicts the limit at least as well (Methods).

Supplementary Note 3 | Robustness of the derived quantities, in full. Because several derived respiratory quantities depend on the estimated starting cell number, we tested which conclusions were sensitive to the inoculum back-projection. Per-cell respiration and carbon-use efficiency depend on the starting cell number N0, obtained by back-projecting the inoculum over the pre-fit interval, $N_0 = N_{\text{inoc}} \cdot \exp(r \cdot \Delta t)$. The specific growth rate r and its activation energy are recovered directly from the oxygen dynamics and are independent of N0; only the per-cell respiratory flux, CUE and quantities derived from them inherit its uncertainty, so we assessed the two ways the back-projection can bias them. First, a fit-window selection issue was corrected: for a small number of series (chiefly at 26 °C) the selector had latched onto a short late-run segment and returned implausible growth rates. These windows were reviewed and corrected to span the full respiration phase, restoring the expected unimodal growth curves and leaving every *C. auris* clade optimum in the 34.8–36.4 °C range.

Second, and more consequentially, the vials are inoculated near bench temperature and take tens of minutes to reach setpoint, so the growth accrued during the pre-fit interval is not governed by the stable r used in the back-projection. Using the continuously logged internal vial temperature together with each clade's own measured growth-temperature curve, we recomputed the growth actually accrued during equilibration and propagated the difference into N_0 . Growth per cell is unaffected by construction. Per-cell respiration and CUE shift in a temperature-structured way: negligibly across 22–36 °C, then increasingly toward the extremes, with the sign set by the growth optimum (below it, a cooler onset implies slower early growth and slightly higher per-cell respiration; above it, the reverse). The median shift on respiration reaches about +14% near 30 °C and –27% at 44 °C, mirrored and buffered in CUE, while growth per cell is immune throughout (Supplementary Fig. 7). Shown on the thermal curves themselves, widening each posterior credible interval by this component leaves an outer envelope that coincides with the credible interval through the mid-range and separates only toward the extremes (Supplementary Figs. 8 and 9).

This equilibration correction is roughly an order of magnitude smaller than the trends it could threaten: the CUE optimum still lies below 37 °C in every taxon, growth remains credibly more temperature-sensitive than respiration, and respiration's monotonic rise is unchanged. We therefore keep the per-cell point estimates unchanged but treat their absolute values at the temperature extremes (≥ 40 °C) as approximate.

A stronger stress test is to remove the back-projection term entirely ($N_0 = N_{\text{inoc}}$), the opposite bound to the equilibration correction. This is not the physical null (the inoculum did grow before the fit window, which is what the back-projection accounts for), so the true values lie between it and the published table rather than at either end; but it isolates every quantity that leans on the term. The primary conclusions are unchanged or strengthened under it: r and its activation energy are independent of N_0 by construction, and the sub-37 °C CUE optimum in fact sharpens. Two secondary results are sensitive and we treat them accordingly. First, the fine fever-cost ordering among the *C. auris* clades: its leader moves from first to fourth of five when the term is removed, whereas the *C. auris* versus *C. parapsilosis* contrast survives both bounds; the term-free analysis sharpens this further, collapsing the between-clade spread in respiration activation energy from 0.22 eV (with the term) to 0.07 eV, so we do not report a within-*auris* ranking at all. The between-clade respiration activation energy under all three treatments, with the term (published), equilibration-corrected, and term-free, is compared directly in Supplementary Fig. 10.

A separate question is whether the manual choice of fit window matters at all. We refitted the same hierarchical growth model on the same traces with windows chosen automatically by the pre-registered classifier's rule instead, so that the two arms differ only in how the interval was selected. Every group's fitted optimum moves down, by 0.45 to 1.26 °C, and every interval includes zero (posterior probability of a positive shift 0.17 to 0.37); activation energies move by at most 0.19 eV, and the deactivation midpoint by at most 0.83 °C except in *C. auris* II, the group that loses most data under the automatic rule. The ordering of groups is preserved except for *C. parapsilosis* and *C. duobushaemulonii*, whose optima lie 0.2 °C apart and which exchange places (Supplementary Fig. 11). No conclusion in this paper turns on either the direction or the magnitude of that shift. The comparison is internally consistent because both arms use identical priors; its manual arm reproduces the published optima with a systematic offset of +0.24 to +0.64 °C, which we traced to prior differences between the refit script and the analysis pipeline rather than to the data.

Supplementary Note 4 | High-oxygen refit. Measured oxygen at the manual window start was 6.5 to 12.9 mg L⁻¹ (median 9.4) and at the window end a median 0.51 of the start (5th to 95th percentile 0.32 to 0.94), so the primary windows already use the oxygen-rich half of each trace. Truncating every window at 50% of its start value shortened 592 of the 1,222 windows, by a median 6%, and left r and K in those wells within 2% of the primary values (median ratios 1.015 and 0.983). The model-free CUE optimum from the truncated fits is within 0.1 °C of the primary value in every group (27.8, 34.0, 31.2, 32.0, 26.5, 28.5 and 31.3 °C for *C. auris* Clades I to IV, *C. parapsilosis*, *C. duobushaemulonii* and *C. haemulonii*), and the Arrhenius activation energy of K over 22 to 38 °C moves by at most 0.05 eV (reprocess/hi02/manual_arms_comparison.csv). The per-well likelihood model rerun on traces truncated at the same cutoff leaves 1,179 of 1,200 wells identifiable and agrees with the full-trace growth call in 1,127 of them, with the 40 °C contrast unchanged (58 of 59 identifiable *C. auris* wells against 11 of 40 relatives; reprocess/hi02/lik_hi02_0.50.csv).

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